

RESEARCH ARTICLE

Explainability-Based Optimized Deep Learning in Histopathological Diagnosis of Multiple Cancers and Development of Mobile Application

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ABSTRACT

Histopathological image analysis is critical for cancer diagnosis, yet many existing models suffer from limited interpretability, high computational demands, and suboptimal classification accuracy. To overcome these limitations, we propose a novel model, Complementary Residual Retentive Network with Guided Gaussian Combined Arms Algorithm (C2RN2GC2A), designed to enhance efficiency and accuracy in cancer classification from histopathological images. C2RN2GC2A is a deep learning model that assimilates residual learning with optimized Gaussian perturbations, thus enhancing both feature extraction and working time in classification tasks. The system merges 2GC2A, a metaheuristic optimization approach motivated by military tactics, for the purpose of improving feature selection, truncating training loss, and speeding up convergence. The Two-stage Guided Chaotic Capuchin Algorithm (2GC2A) brings together Gaussian perturbations with a combined arms tactic, which makes it possible to do a good job of both exploring and exploiting the search space for better parameter tuning. In order to achieve interpretability, Layer-wise DeepLIFT Relevance Propagation (LDLRP) is used to delineate the significant areas of the image that have an impact on the classification, thus making the process more transparent and building up clinical trust. LDLRP is a cutting-edge explainable AI technology that grants relevance ratings to input characteristics and thus allows the model to visually demonstrate the most significant regions in histopathological images, thereby facilitating clinical decision-making. Testing on LC25000 produced a remarkable accuracy of 98.02% along with a minuscule training loss of 0.08, besides which there were 13 false positives and 29 false negatives. On BreakHis, the accuracy was 98.54%, and the validation loss was 0.05, with 98 false positives and 112 false negatives. The proposed framework significantly improves diagnostic reliability, classification accuracy, and clinical transparency in multi-cancer histopathological image analysis.

1 | Introduction

Histopathological analysis, which is the microscopic examination of tissues, remains the gold standard for recognizing various cancers by uncovering the malignant features and monitoring the disease's evolution (Istrate-Ofițeru

et al. 2024). Although it is very important, the manual interpretation of histopathological images by pathologists is a slow process, is dependent on different people's interpretation, and is inconsistent both within the same observer and between different observers (Moscalu et al. 2023). The use of such methods often results in delayed diagnoses and

Summary

- Developed C2RN2GC2A for accurate multi-cancer image classification.
- Introduced LDLRP to visualize decisions and assess disease severity.
- Used augmentation to improve data diversity and model generalization.
- Employed MPKCNN for effective segmentation of complex image structures.

inconsistencies that negatively impact patients' health. Deep Learning (DL), which is one of the branches of Artificial Intelligence (AI), has emerged in the gantry for automating and enhancing medical image analysis in recent years (Dos Santos et al. 2023). Convolutional Neural Networks (CNNs), especially, have proved to be extraordinarily able to learn complex patterns from large histopathological datasets, thus being able to accurately detect, segment, and classify cancerous tissues (Shi et al. 2023). By leveraging hierarchical feature extraction, these models identify subtle morphological cues that are often challenging for human observers to discern (Wang et al. 2023; Wen et al. 2023). The application of explainability methods makes the clinicians trust the models more and helps them to understand the predictions made by the models. This collaboration of DL and histopathology is expected to change the cancer diagnosis face, making it possible to have faster diagnoses, tailored therapy plans, and overall improved patient care (Munnangi and Satheeskumar 2025). Starting from such a strong base, lately, the evolution of DL has resulted in the introduction of several very powerful models that have significantly improved the accuracy and speed of histopathological analysis. The Self-Organized operational Neural Network (Self-ONN) learned very efficiently complex tissue structures for the recognition of lung and colon cancer, which in turn stimulated the development of representation learning (Said et al. 2024). Transfer learning based on EfficientNet-B3 used pretrained weights and showed high generalization and classification accuracy (Ochoa-Ornelas et al. 2025). The Compact Convolutional Neural Network (CCNN) proposed by Attallah (2025) was a fusion of multiscale deep features and feature selection, which helped in classifying images at different resolutions very robustly. Besides, the Lightweight Efficient Vision Transformer (LEViT) promoted cross-modality learning between Computed Tomography (CT) and histopathology, which led to the diagnosis of lung cancer (Mahmud et al. 2025). ColonNet, a heteromorphous CNN, ensured reliable tumor detection in colon and lung tissues via shape-sensitive feature extraction (Iqbal et al. 2023). In order to improve the robustness of the ResNet101 version, it was chosen to be retrained on the cancer classification project, and thus the model could deal with different scenarios (El-Ghany et al. 2023). Conversely, DeepHistoNet, through the combination of deep features, enabled the multi-class classification of tumors in liver, lung, colon, and a plethora of anatomically modern and extinct human (hominine) samples (Kadirappa et al. 2024). DenseNet169, which is recognized as one of the most effective AI-based breast cancer detection systems, not only includes explicable AI but also does so via fine-tuned

reconstruction (Talukder 2025). EfficientNet-B2 with supervised contrastive learning was the one to deal with class imbalance in binary classification problems (Sani et al. 2025). The application of Quantized Scale-Local Binary Patterns (QS-LBP) along with CNN has facilitated the extraction of the texture details required for the diagnosis of breast cancer (Gül 2025). Deep Neural Network-eXtreme gradient Boosting (DNN-XBoost) done at the Indian Institute of Technology, Madras, has improved interpretability and reliability through hybrid learning (Maleki et al. 2023), while the Brea-Net model used the dual-attention mechanisms to support its strong classification performance on imbalanced datasets (Liang and Meng 2023). Furthermore, the application of DL and machine learning ensemble approaches for early detection of lung cancer was demonstrated, with high accuracy achieved by combining the two methods (Ansari et al. 2025). One of the subsequent improvements is the use of machine learning modeling for the identification of prognostic hub genes in cervical adenocarcinoma, and consequently, this is expected to provide better surveillance for immunotherapy and lead to the creation of multi-target treatments with improved clinical outcomes (Abbasi, Abbasi, et al. 2025; Abbasi, Amin, et al. 2025). In the field of ophthalmology, the application of Adaptive Deep Convolutional Neural Networks (ADCNN) has been beneficial in the detection of diabetic retinopathy in fundus images with higher accuracy, and therefore, the increase in the accuracy of the diagnosis has been achieved (Abbasi, Abbasi, et al. 2025; Abbasi, Amin, et al. 2025). Besides this, the research on secure communication protocols has resulted in the development of communication methods that are lossless and efficient for the healthcare systems of smart cities, especially in 6G Internet-of-Things (IoT) settings, not only guaranteeing reliable data interchange for the healthcare applications but also (Abbasi et al. 2024). All these models together pushed the technique toward automated, accurate, and interpretable cancer diagnosis.

1.1 | Research Gaps

DL has shown remarkable potential in histopathological cancer diagnosis, but current models often lack interpretability, robustness, and generalizability. To bridge these gaps, the proposed C2RN2GC2A model integrates optimized learning and explainability mechanisms to enhance multi-class classification accuracy and clinical trust, addressing the research gaps and the challenges shown in Figure 1.

This research, which introduces the C2RN2GC2A model for cancer classification using histopathological images, is guided by three key research questions:

1. How can the unified DL architecture be developed to deliver consistent performance across multiple cancer classification tasks using diverse histopathological datasets?
2. To what extent have recent explainability AI models contributed to the interpretation of cancer diagnosis models, utilizing, in the context of LDLRDs?
3. How does the combination of MPKCNN together with the use of advanced data augmentation methods improve

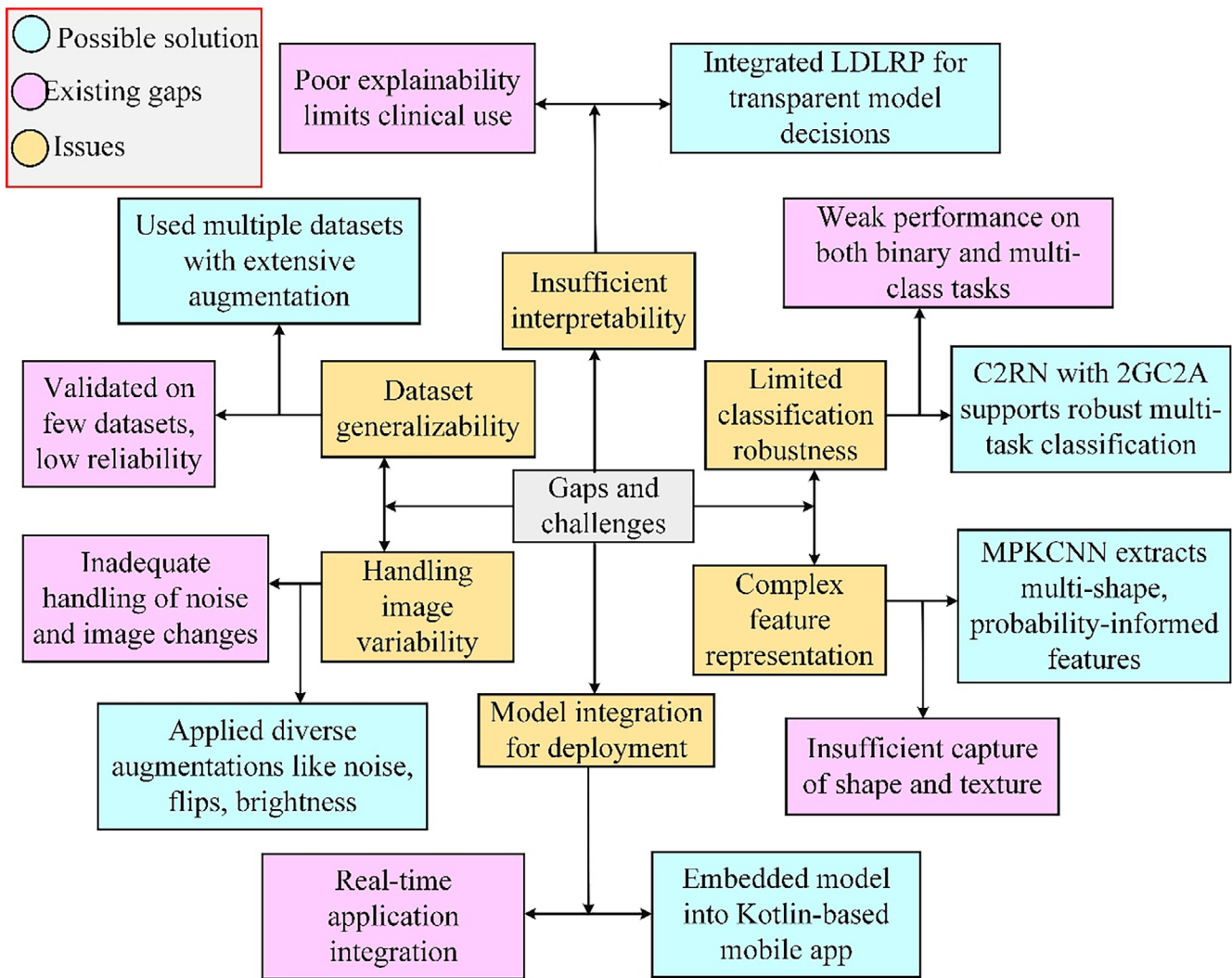


FIGURE 1 | Research gaps and challenges.

the generalizability and applicability of histopathological image classification models, mainly in the area of mobile health applications?

1.2 | Research Contributions

1. This research presents a new DL model called C2RN2GC2A, which is suitable for both binary and multi-class classification of breast, colon, and lung cancers through histopathological images. By utilizing complementary residual connections, the model does a better job in deep feature extraction. These connections not only boost feature retention but also improve the overall accuracy of diagnostics, especially in the case of difficult medical image patterns in image analysis.
2. This research proposes a new explainable AI method, LDLRP, which serves as a tool for the interpretability assessment of the cancer classification model. By marking the parts of histopathological images that are decisive for the model's decision, LDLRP offers visual insights that subsequently enhance transparency, trust in the clinic, and diagnostics that are more informed.

3. The study utilizes cutting-edge data augmentation methods to not only multiply the number of different images but also to check the generalization of the model. The images produced by augmentation are then fed into MPKCNN, which is a new convolutional neural network tailored for segmentation, thus enhancing the learning of characteristics from different image structures and resulting in strong and precise cancer classification performance.

1.3 | Research Organization

The structure of this research is organized as follows: Section 2 describes the materials and methodologies employed; Section 3 showcases the results along with their analysis; Section 4 provides a comprehensive discussion reflecting on the overall findings; and Section 5 offers the concluding remarks.

2 | Materials and Methods

The proposed methodology follows a comprehensive, multi-stage pipeline for robust and interpretable histopathological image classification, shown in Figure 2. As a first step, input data

coming from two benchmark datasets, for instance, LC25000 and BreakHis, are consumed to assess the performance of the model in five-class and binary cancer classification tasks, respectively. To fortify strength and generalization, the images are

subjected to various data augmentations, such as the addition of Gaussian noise, modulation of brightness and contrast, flipping, rotating, and color jittering. Afterward, the augmented images are fed to the novel MPKCNN, which isolates the significant

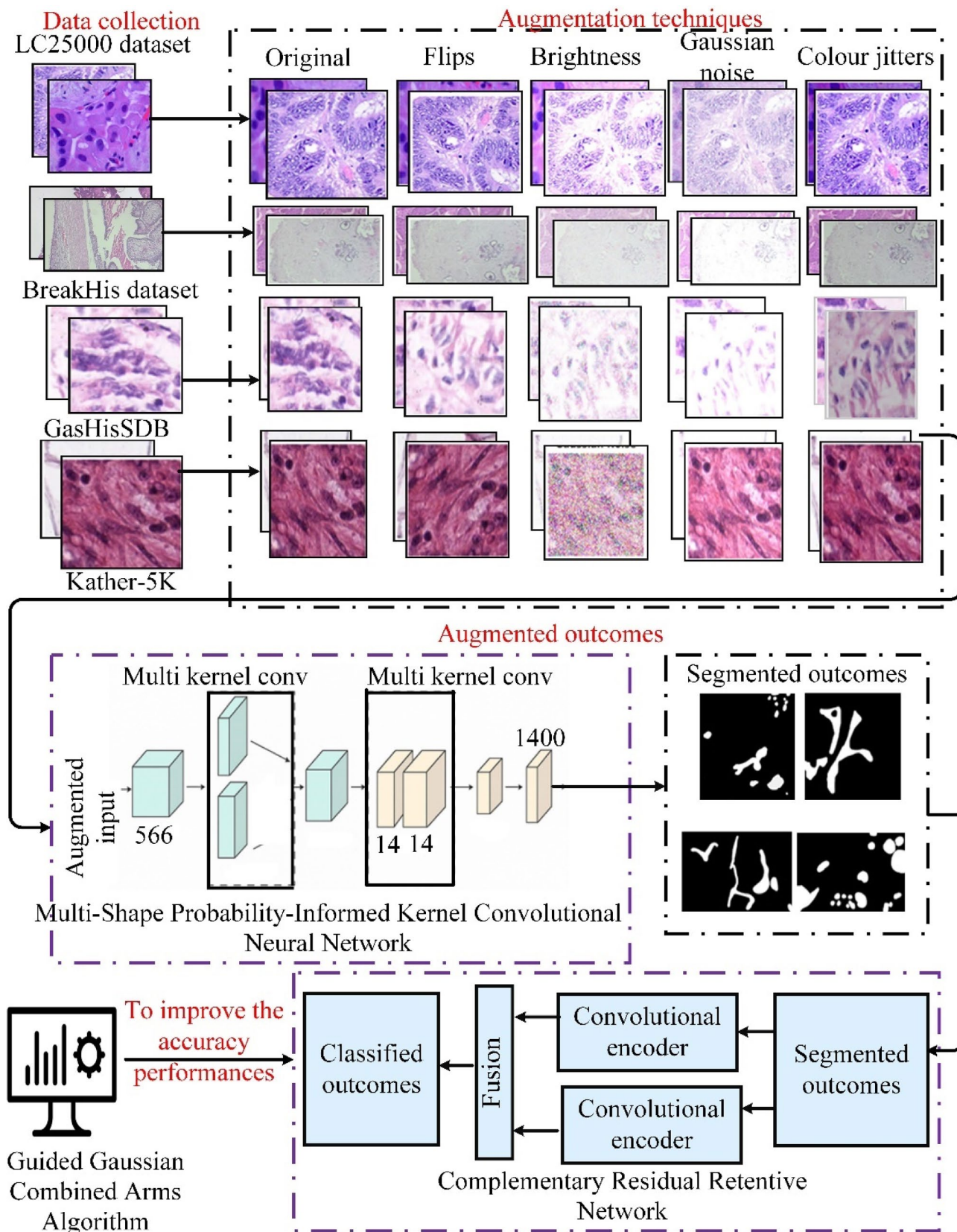


FIGURE 2 | Workflow architecture of the introduced model.

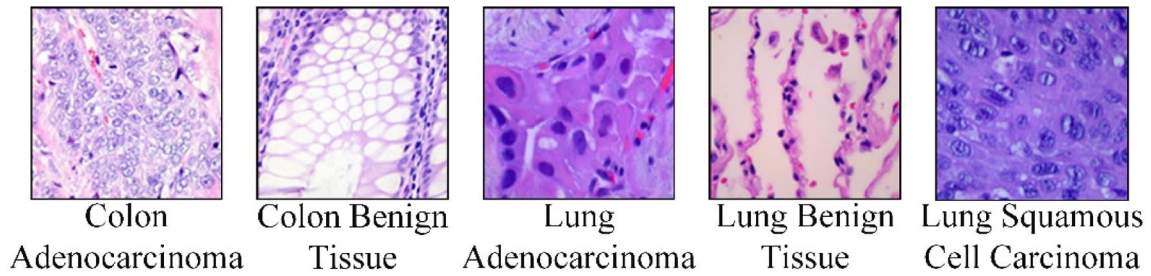


FIGURE 3 | Sample class of the LC25000 dataset.

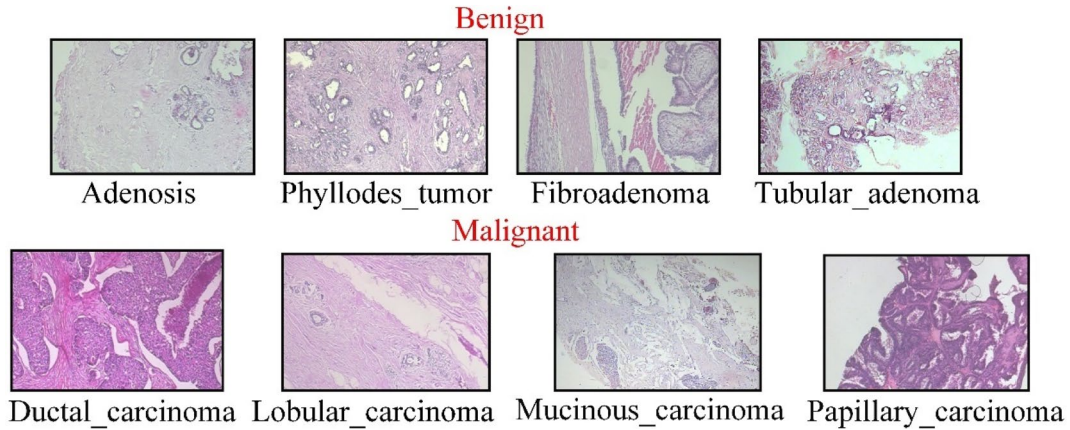


FIGURE 4 | Sample images of the BreakHis dataset.

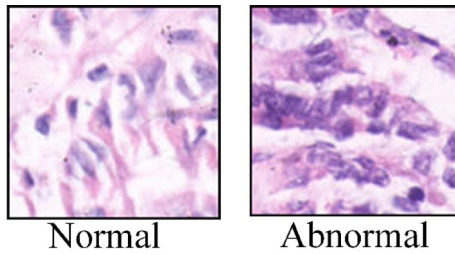


FIGURE 5 | Sample images of the GasHisSDB dataset.

areas of interest by using probabilistic spatial kernels that are formed to fit the different lesion shapes. The features that have been segmented are given to the C2RN, which has a 2GC2A incorporated for the best parameter tuning and efficient classification. The C2RN structure takes advantage of multi-scale residual blocks with attention-conscious skip connections to improve the discriminative learning while in the process, reducing information loss. The 2GC2A optimizer employs adaptive Gaussian perturbations and collaborative solution updates to accelerate convergence and to avoid getting trapped in local optima.

Finally, interpretability and clinical relevance are ensured using the LDLRP technique, which visualizes the model's decision-making mechanism, clarifying it and facilitating the evaluation of disease severity through relevance score maps, thereby enabling more transparent and clinically reliable outcomes.

2.1 | Data Collection

The LC25000 dataset (Said et al. 2024) comprises 25,000 histological images obtained from biopsy samples to diagnose and

differentiate early-stage lung and colon cancer. Figure 3 provides example images from various classes within the LC25000 dataset.

The dataset is made up of five classes that are equally distributed: Colon Adenocarcinoma (5000), Colon Benign Tissue (5000), Lung Adenocarcinoma (5000), Lung Benign Tissue (5000), and Lung Squamous Cell Carcinoma (5000). The natural images' initial spatial resolution was 1024×768 , which got cropped to 768×768 prior to the data augmentation process. The BreakHis dataset (Gül 2025) includes a total of 7909 stained microscopic images of breast tumor tissue derived from 82 patients, with pictures taken at four different magnification levels: 40×, 100×, 200×, and 400×. The various classes of the BreakHis dataset are visually represented in Figure 4.

It includes two main categories: benign (2480 images) and malignant (5429 images). The dataset is imbalanced, with a higher number of malignant samples.

GasHisSDB (Solmaz and Tasci 2025) consists of a total of 245,196 images of histopathological gastric cancer from 600 slides of tissue samples taken at Longhua Hospital, Shanghai University of Traditional Chinese Medicine, which contain 97,076 abnormal and 148,120 normal images. The various-sized sample images are illustrated in Figure 5.

The Kather-5K dataset (Liu et al. 2024) contains 5000 patch images categorized into 8 groups in the following distribution: images of tumor epithelium consist of 1000 images; simple stroma contains 500 images; homogeneous tissue, which was defined as tumoral stroma, consisted of 1000 images; 500 images of muscle tissue; 500 images of complex stroma; 500

images of immune cells, or conglomerates of immune cells; 250 images of debris; and 250 images of necrosis, or mucus; 500 images are of normal colon mucosa; 250 images of adipose tissue; and 250 images of background tissue, and the sample images are also added in Figure 6. The equal distribution of data serves as a remarkable base for the training and evaluation of histopathological image analysis models. The division of data for every dataset is usually done into two parts, one being training (70%) and the other testing (30%), so that the model is trained and evaluated on different data. A weighted loss function is used during the training process to improve the already existing classification performance, and thus, this issue of class imbalance is tackled.

2.2 | Data Augmentation

Data augmentation is applied to the dataset images, which enhances the model's generalization and robustness against the different imaging conditions and thus reduces the risk of overfitting (Mahajan and Singla 2024). To replicate the noise from the real world, Gaussian noise is applied to the images, thus creating pixel-level differences as illustrated by Equation (1):

$$Q_{GN} = Q(a, b) + N(0, \sigma^2) \quad (1)$$

where $Q(a, b)$ denotes the intensity of the original image at pixel location (a, b) , and $N(0, \sigma^2)$ represents Gaussian noise with zero mean and variance σ^2 . Next, brightness and contrast adjustments are performed to replicate changes in lighting conditions. The augmented pixel intensity is calculated as in Equation (2):

$$Q_{bc}(a, b) = \alpha \cdot Q(a, b) + \beta \quad (2)$$

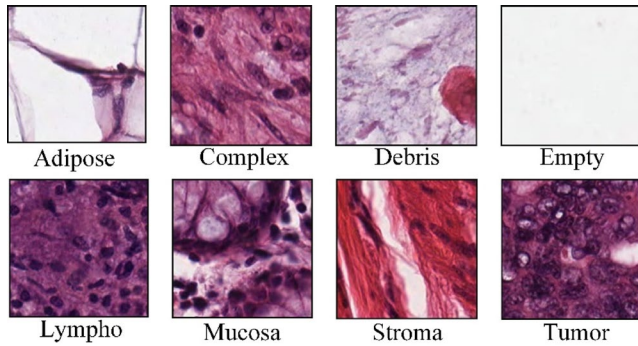


FIGURE 6 | Sample images of the Kather-5K dataset.

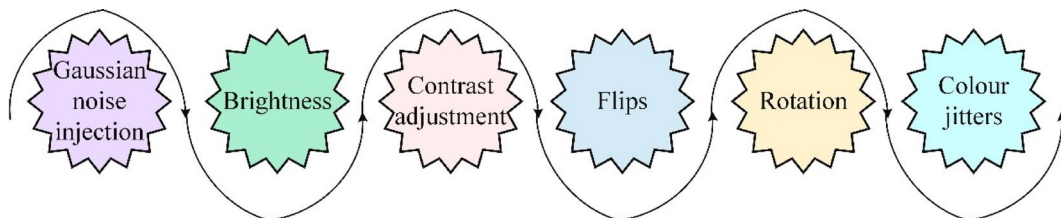


FIGURE 7 | Augmentation process.

where α controls the contrast and β modifies the brightness. In addition, geometric transformations are incorporated, including random horizontal and vertical flips and rotations shown in Figure 7.

The rotation transformation is applied using the rotation matrix, using equation (3):

$$\begin{bmatrix} a' \\ b' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} \quad (3)$$

where θ is the randomly selected rotation angle, and (a', b') is the new coordinate after rotation. The application of color jittering increases variability through the random alteration of brightness, contrast, saturation, and hue, and this, in turn, makes it more robust against color inconsistencies. Images augmented this way are better in learning features and generalizing, and then they are sent to the segmentation module for precise region delineation.

2.3 | Multi-Shape Probability-Informed Kernel Convolutional Neural Segmentation for Histological Images

Post-augmentation, the histological images are inputted into MPKCNN, which is a new model that enhances the segmentation of difficult-to-interpret biomedical images through the combination of multi-shaped convolutional kernels and a probability-guided attention mechanism (Guo et al. 2024; Reddy and Soma 2025). The intricate structure of this model is shown in Figure 8.

Unlike standard CNNs that apply uniform kernels, MPKCNN employs a collection of diverse kernel shapes (e.g., rectangular, circular, and diagonal) to more effectively extract spatially discriminative features. For each convolutional layer l , the segmented feature map $F^{(l)}$ at pixel location (i, j) is computed using equation (4):

$$F_{ij}^{(l)} = \sum_{s=1}^S \sum_{(m,n) \in \Omega_s} K_{m,n}^{(s)} \cdot I_{i+m,j+n}^{(l-1)} \quad (4)$$

where s denotes the number of kernel shapes, Ω_s defines the receptive field corresponding to the shape s , $K^{(s)}$ is the shape-specific kernel, and represents the input feature map from the previous layer. To enhance contextual awareness, MPKCNN incorporates a probability-informed attention map P , which assigns dynamic importance to each spatial location based on the

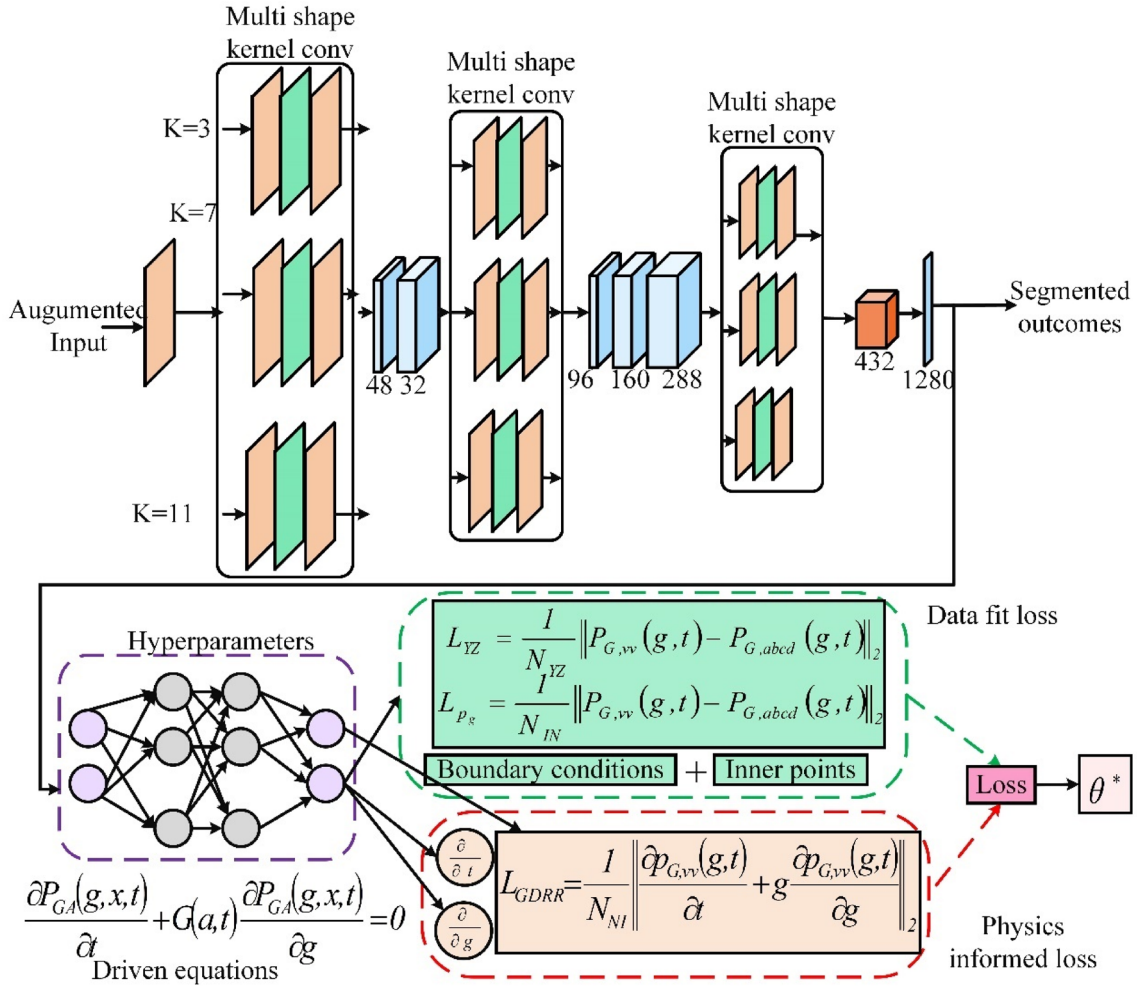


FIGURE 8 | Architecture of the MPKCNN model.

likelihood of tumor presence. This mechanism modulates the convolutional operation as in equation (5):

$$F_{i,j}^{(l)} = \sum_{s=1}^S \sum_{(m,n) \in \Omega_s} P_{i+m,j+n}^{(l-1)} \cdot K_{m,n}^{(s)} \cdot I_{i+m,j+n}^{(l-1)} \quad (5)$$

where $P_{i+m,j+n}^{(l-1)}$ denotes the probability attention map generated by a SoftMax activation applied to intermediate activations, enabling the model to focus on high-probability regions. The two-fold approach of multi-shape adaptability and probability-assisted filtering allows MPKCNN to reach very precise and restricted segmentation, which is vital for the next diagnostic activities, such as tumor classification and grading. The upscale feature maps, which are refined on the basis of the probability calculations produced by the MPKCNN model, are directed toward a dedicated classification network.

2.4 | Histopathological Image Classification With Complementary Residual Retentive Network

After segmentation, extracted and refined region-based features are input to the Complementary Residual Retentive Network (CRRN) (Wilburn et al. 2023). CRRN achieves precise categorization

through its residual learning feature retention mechanisms. The innovative architecture of CRRN is formed by combining the deep residual connections along with the attention-based segments; the latter are designed to excellently take care of the morphological diversity among the different histopathological cancer tissue types. It integrates deep residual connections and attention-based retention modules to mean the morphological diversity in-between histopathological cancer tissue types. Specifically, the input features $X \in \mathbb{R}^{H \times W \times C}$ from the segmentation stage are processed through a series of Complementary Residual Retentive (CRR) modules, which combine traditional residual learning with an additional feature retention mechanism.

In contrast to conventional ResNets and DenseNets that mainly utilize deep residual connections or dense concatenation of features, CRRN utilizes spatial and channel-wise attention as a filter for the significant features in every layer. Thus, the combination of the mechanisms leads CRRN to keep the important pathological patterns and, at the same time, to let the gradients flow through the deep layers with less resistance. The formulation of the CRR module is given by Equation (6):

$$B^{(l)} = A^{(l)} + F(A^{(l)}; \theta^{(l)}) + R(A^{(l)}; \varphi^{(l)}) \quad (6)$$

where F signifies the residual mapping, and R is the retention function applies spatial and channel-wise attention to emphasize salient features. Subsequently, features from multiple CRR modules are aggregated via a multi-scale fusion process using Equation (7):

$$F_{fused} = M(B^{(1)}, Y^{(2)}, \dots, Y^{(L)}) \quad (7)$$

which integrates hierarchical representations to represent different tissue structures at different scales. This multi-scale fusion greatly enhances the model's capacity to manage the complex variations in tissue morphology, as opposed to the conventional methods, which frequently encounter such complexity as a challenge. The fused feature vector F_{fused} is subsequently sent through fully connected layers, ending with a SoftMax classifier to yield class probabilities that are predicted and represented through Equation (8):

$$\hat{b} = \text{Soft max} (W_c \cdot \text{Flatten}(F_{fused}) + b_c) \quad (8)$$

where $\hat{b}$ denotes the probability distribution across the defined cancer types, thus allowing very accurate binary and multi-class separation. Table 1 provides a comprehensive strategy for the C2RN2GC2A technique, whereby starting from the input pre-processing, it goes through the final classification stages. The inclusion of MPKCNN, complementary residual modules, and explainability mechanisms is also depicted.

This integrated segmentation-classification framework effectively combines region-focused feature extraction and deep contextual representation learning, thereby enhancing the model's ability to exactly diagnose and differentiate breast, colon, and lung cancer subtypes in histopathological images. The inclusion of the complementary attention mechanism in CRRN offers a more targeted focus on significant features, which is a departure from the purely residual or concatenation-based approaches seen in ResNet and DenseNet architectures.

2.5 | Loss Function and Training Strategy

To optimize CRRN, a weighted cross-entropy loss is employed, accommodating class imbalance typical in medical datasets, which is expressed using Equation (9):

$$L = - \sum_{k=1}^K w_k \cdot b_k \log \hat{b}_k \quad (9)$$

where b_k is the ground-truth label, $\hat{b}_k$ is the predicted probability for the class k , and w_k is the weight inversely proportional to class frequency, ensuring that the learning focus is adequate for minority classes. The weighted cross-entropy loss is minimized, which leads to learning, and in parallel, an advanced meta-heuristic optimizer improves the optimization efficiency, thus avoiding local minima during training.

2.6 | Parameter Optimization Using Guided Gaussian Combined Arms Algorithm

To further enhance the optimization process of the network parameters and avoid local minima, this integrates the 2GC2A as an intelligent metaheuristic optimizer (Etesami et al. 2025). The 2GC2A is based on military combined arms strategies and Gaussian perturbations, which make it possible to do the effective exploration and exploitation of the search space.

The 2GC2A begins with an initialization step, where a population of candidate solutions $\{A_i\}_{i=1}^N$ is generated randomly within the feasible parameter bounds. This initialization is mathematically expressed as in Equation (10):

$$A_i^{(0)} = A_{\min} + r_i \times (A_{\max} - A_{\min}), \quad i = 1, 2, \dots, N \quad (10)$$

where $A_{\min}$ and $A_{\max}$ the lower and upper limits of the solution space, respectively, and $r_i \sim U(0, 1)$ is a uniformly distributed

TABLE 1 | The architecture of the proposed C2RN2GC2A model.

Layers	Operations	Input shape	Output shape
Input	Raw histopathological image	$512 \times 512 \times 3$	$512 \times 512 \times 3$
Initial conv layer	7×7 Conv, stride 2 + BatchNorm + ReLU	$512 \times 512 \times 3$	$256 \times 256 \times 64$
Pooling layer	3×3 Max Pool, stride 2	$256 \times 256 \times 64$	$128 \times 128 \times 64$
Residual block (1)	$[1 \times 1 \text{ Conv} \rightarrow 3 \times 3 \text{ Conv}] \times 6$	$128 \times 128 \times 64$	$128 \times 128 \times 256$
Transition layer (1)	$1 \times 1 \text{ Conv} + 2 \times 2 \text{ Avg Pool}$, stride 2	$128 \times 128 \times 256$	$64 \times 64 \times 128$
Residual block (2)	$[1 \times 1 \text{ Conv} \rightarrow 3 \times 3 \text{ Conv}] \times 12$	$64 \times 64 \times 128$	$64 \times 64 \times 512$
Transition layer (2)	$1 \times 1 \text{ Conv} + 2 \times 2 \text{ Avg Pool}$, stride 2	$64 \times 64 \times 512$	$32 \times 32 \times 256$
Complementary block	Multi-path residual + complementary attention fusion	$32 \times 32 \times 256$	$32 \times 32 \times 512$
MPKCNN layer	Multi-shape probability-informed kernel CNN layer	$32 \times 32 \times 512$	$32 \times 32 \times 1024$
Global average pooling	GAP	$32 \times 32 \times 1024$	$1 \times 1 \times 1024$
LDLRP (Explainability)	—	$1 \times 1 \times 1024$	Heatmap Output
Classification	Fully connected + SoftMax	$1 \times 1 \times 1024$	3 or 4 classes

random vector guaranteeing diversity in the initial population. The optimization process of 2GC2A is depicted in Figure 9.

During iterative optimization, each candidate solution's quality is assessed using a fitness function $Fitness\ function(A_i)$, which quantifies the objective of minimizing classification error while maximizing generalization. The fitness function is typically based on classification loss and regularization terms, but in this case, it is primarily focused on the weighted cross-entropy loss obtained from the output of the CRRN, mathematically expressed as in Equation (11):

$$Fitness\ function(A_i) = L(A_i) \quad (11)$$

The optimization process in 2GC2A follows an iterative scheme where candidate solutions are updated based on a Gaussian perturbation and a directional update based on the combined arms strategy. The update rule is shown in Equation (12):

$$A_i^{t+1} = A_i^t + \alpha \cdot (P_i^t + D_i^t) \quad (12)$$

where A_i^t is the candidate solution at iteration t . α is the learning rate (step size). The Gaussian perturbation P_i^t , which introduces randomness to ensure diversity in the population, D_i^t is a source of randomness. The update of the direction is based on the combined arms strategy, which promotes collaboration in exploring the solution space. Mathematically, the Gaussian perturbation P_i^t is defined by Equation (13)

$$P_i^t = N(0, \sigma^2) \quad (13)$$

where $N(0, \sigma^2)$ is a Gaussian distribution with mean 0 and variance σ^2 , helping the algorithm explore unvisited regions of the solution space. The directional update D_i^t is determined by the combined arms strategy, which balances exploitation (searching around the best solutions) and exploration (searching in new regions). This is mathematically described as in Equation (14):

$$D_i^t = \beta \cdot (G_i^t + C_i^t) \quad (14)$$

where G_i^t is the guiding vector from the global best solution to the candidate solution. C_i^t is the collaborative vector, encouraging the exchange of information among candidate solutions. β is the criterion that influences the degree of calculation and

exploration for new answers. The four-step iterative optimization process of 2GC2A eventually runs for the exposure of a stopping criterion, like convergence or a maximum number of iterations, to be reached. On the other hand, the 2GC2A algorithm is totally different from the optimization process by the use of the gradients through the adoption of a non-fully gradient-based meta-heuristic optimization method. Therefore, 2GC2A can be very useful in scenarios of difficult optimization landscapes like the classification of histopathological images, where the nature of the solution space is non-linear, and the use of traditional methods for traversal.

2.7 | Algorithmic Complexity and Optimization Efficiency of 2GC2A

The 2GC2A differs from traditional gradient-based optimizers like Adam or SGD in that it avoids reliance on gradients for parameter updates. While methods like SGD update parameters based on the gradient of the loss function, as shown in the update rule, are expressed as in Equation (15):

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta} J(\theta_t) \quad (15)$$

where θ_t represents the parameters at the time step t , η is the learning rate, and $\nabla_{\theta} J(\theta_t)$ is the gradient of the loss function, the computational complexity of SGD is $O(n \cdot d)$ where n is the number of data points and d is the dimensionality of the parameter space. Similarly, Adam improves on SGD by computing adaptive learning rates using moment estimates. Adam's update rule is expressed as in Equation (16):

$$\theta_{t+1} = \theta_t - \frac{\eta}{\sqrt{\hat{v}_t + \epsilon}} \hat{m}_t \quad (16)$$

Adam's complexity remains $O(n \cdot d)$ due to the extra computation for moment estimation. On the other hand, 2GC2A employs a metaheuristic optimization method that mixes random exploration and strategic exploitation. At every iteration, it applies Gaussian perturbations to candidate solutions in order to investigate the search space, as demonstrated in Equation (17):

$$\theta_{t+1} = \theta_t + N(0, \sigma^2) \quad (17)$$

where $N(0, \sigma^2)$ denotes Gaussian noise with zero mean and variance σ^2 . This lets 2GC2A effectively escape local minima and cultivate awareness of wider regions of the solution space without relying on gradients. The exploitation phase refines candidate solutions based on the best-found solution, using an update rule like is shown in Equation (18):

$$\theta_{t+1} = \theta_t + \alpha \cdot (\theta_{best} - \theta_t) \quad (18)$$

where θ_{best} is the best solution found so far, and α is a scaling factor. This step focuses on converging the search toward the optimal region of the solution space. The overall computational complexity of 2GC2A per iteration is $O(n)$, where n is the number of candidate solutions. The reason for this complexity is that during the process of each iteration, the method 2GC2A only embodies the application of Gaussian perturbations and the evaluation of the fitness of

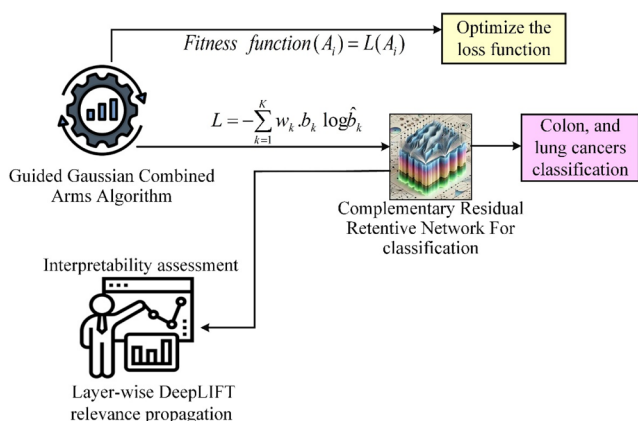


FIGURE 9 | Overview of 2GC2A optimization.

the candidate solutions, which are both linear operations $O(n \cdot d)$. By not doing gradient computation and applying the exploration-exploitation strategy, 2GC2A turns out to be more efficient in high-dimensional spaces as compared to Adam and SGD, which both have computational complexity $O(n)$. One of the advantages of 2GC2A's linear complexity is that it is able to scale well with the number of candidates, thus making it very suitable for those applications that require working with large and complex datasets, where traditional optimizers face challenges because of the high dimensionality and the risk of getting trapped in local optima. After the optimization process, the interpretability of the model is evaluated with the help of an explainable AI technique that aims to increase transparency and clinical trust.

2.8 | Interpretability Assessment Using Layer-Wise DeepLIFT Relevance Propagation

After optimizing the model with 2GC2A, an interpretability model as LDLRP is developed, which explains predictions by attributing relevance scores to input features, enhancing clinical trust and transparency. The LDLRP method decomposes the model's output by attributing feature-wise relevance indicators, thus revealing which regions of the histopathological image most influenced the classification decision (Ullah et al. 2021; Li et al. 2022). Overview of the LDLRP model is shown in Figure 10.

Given a deep neural network f with input a and output $f(a)$, LDLRP computes the relevance score R_i for each input neuron a_i by propagating the prediction backward through the layers, while preserving the total relevance at each stage. Formally, relevance propagation follows the conservation rule is defined in Equation (19):

$$\sum_i R_i^{(l)} = \sum_j R_j^{(l+1)} \quad (19)$$

where $R_i^{(l)}$ is the relevance attributed to the neuron i in layer l , and $\sum_j R_j^{(l+1)}$ is the relevance at the subsequent layer $l + 1$. LDLRP enhances this framework by integrating DeepLIFT principles

that consider the difference between the activation of each neuron and a reference baseline a_{ref} . The relevance R_i for neuron i is computed as in Equation (20):

$$R_i = (a_i - a_{ref,i}) \times \frac{\partial f(a)}{\partial a_i} \quad (20)$$

Equation (13) quantifies the contributions of the deviations from the baseline input to the final output, providing a very detailed and easy-to-understand explanation of the areas that the model focuses on. LDLRP method utilizes a layer-by-layer approach for the propagation of relevance, which is different from the conventional DeepLIFT method, and consequently, it ensures that the relevance is shifted back through each layer separately, and in this way, it provides comprehensive information about the input layer's participation. This, in turn, increases interpretability, especially in deep architectures, as it reveals the exact features that had an impact on the model's decision. On the other hand, standard DeepLIFT captures the overall importance across layers, thus not having this stage-wise separation. Other methods of propagation, such as Shapley Additive Explanations (SHAP), Local Score-CAM, and Gradient-weighted Class Activation Mapping (Grad-CAM), also bring forth explanations, but each uses different means. SHAP is using the game-theoretic framework to determine the global feature importance; on the other hand, LIME is providing local explanations for single predictions. Grad-CAM is producing visual heatmaps for CNNs, but it does not have the layer-specific insights that LDLRP delivers. By providing a more detailed and precise interpretability of each layer, LDLRP has an edge over deep models, which proves to be of great help in complex scenarios such as medical image analysis.

With the help of LDLRP not only does the model outline the most important histological features that are related to various severities of breast, colon, and lung cancers but also does it in a very systematic way. These visual rationales not only confirm the model's decision-making process but also help pathologists recognize important morphological markers, thus connecting automated analysis and expert interpretation.

2.9 | Development of an On-Device Mobile Application for Real-Time Cancer Detection

A Kotlin-based mobile application for the Android platform has been created to detect cancer in real-time directly on the device by analyzing histopathological images. The app is built upon DL models, the MPKCNN and C2RN2GC2A, which have undergone training on the benchmark datasets and subsequently converted to TensorFlow Lite format for using it effectively offline. The users are provided with the functionality of either capturing or uploading the images, and in either case, the images will be preprocessed to fit the model's input requirements. The application is capable of conducting both binary as well as multi-class classification, in which it mentions not only the presence of cancer but also the type, such as lung, colon, and breast. As a means of gaining clinical trust, it uses LDLRP explainability, which creates heatmaps that show the parts of the image that are informative. A simple, material design interface is used to present the results and interpretability maps with the possibility of saving or

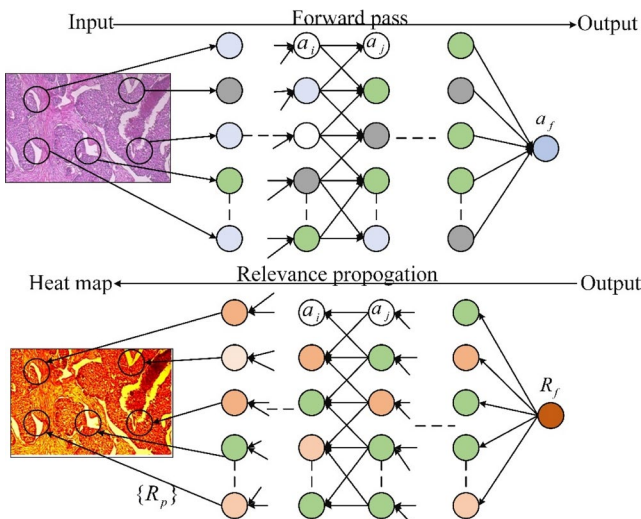


FIGURE 10 | Overview of LDLRP model.

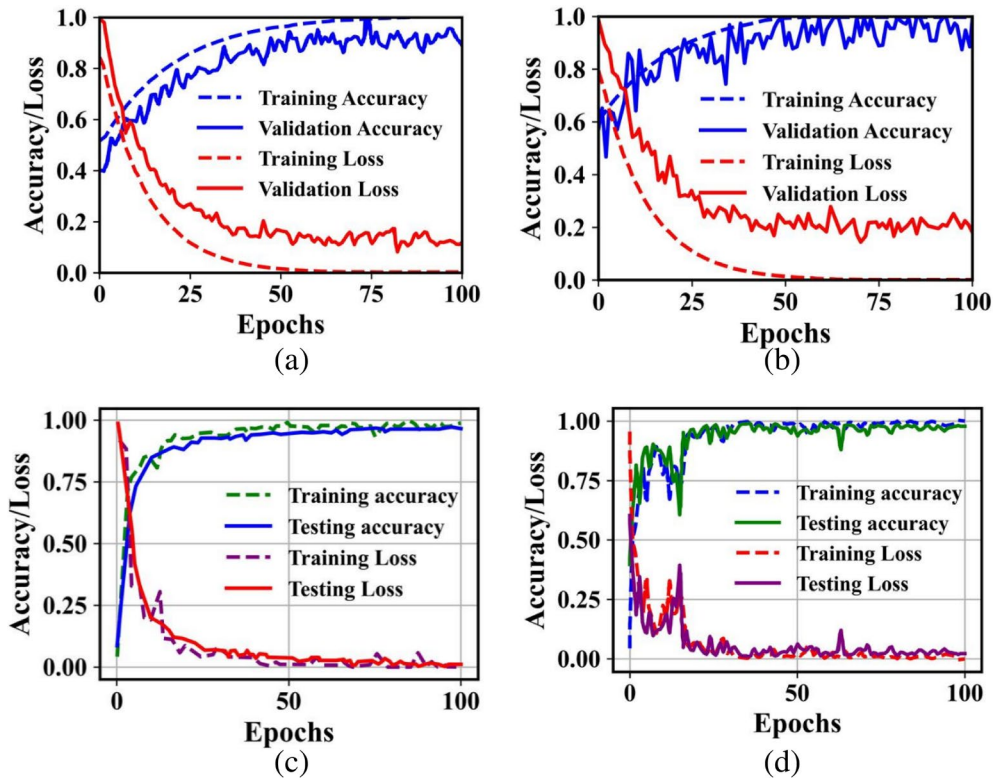


FIGURE 11 | Performance of training and validation of (a) LC25000 and (b) BreakHis (c) GasHisSDB, and (d) Kather-5 K dataset.

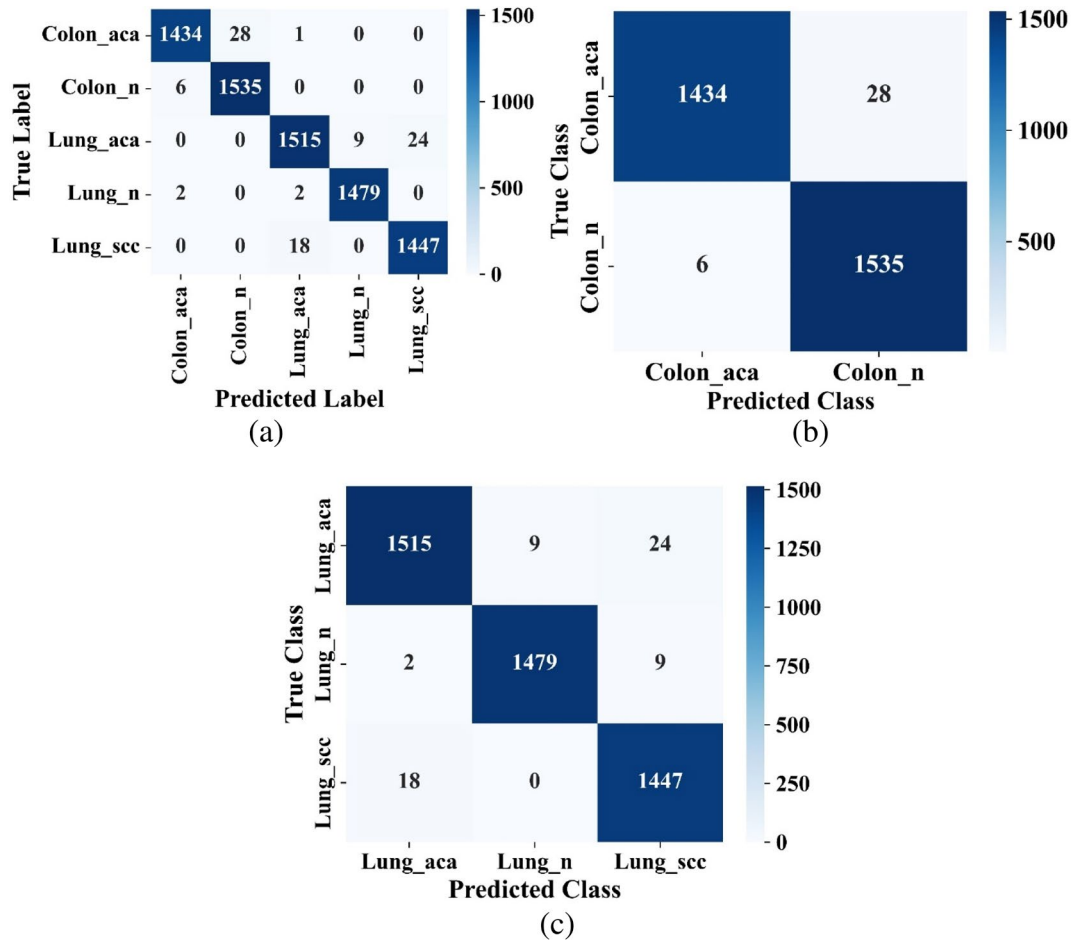


FIGURE 12 | Confusion matrices for (a) Colon and Lung Cancer, (b) Lung Cancer, and (c) Colon Cancer, using the full LC25000 dataset.

sharing the findings, thus encouraging telemedicine. This technology deployment makes it possible to have a mobile, internet-free diagnostic tool that can be used in poor and remote areas.

3 | Results and Discussion

This research is implemented using Python, and the mobile application is developed using the Kotlin programming language. The model's hyperparameters are set as follows: batch size of 64, learning rate of 0.001, dropout rate of 0.2, Adam optimizer, input shape of 512×512 , 100 epochs, and 40% dropout rate. The Softmax activation function is applied in the output layer. The proficient settings are very much like the ones done, which lead to the training being very efficient and overfitting being excessive. The model demonstrates its potential by always delivering reliable results in a variety of histopathological image classification tasks. The setup of the computer is one of the main factors that contribute to the accuracy of the measurements of FLOPs and Inference Time (ms). A high-end NVIDIA Tesla A100 GPU with 40 GB of VRAM was utilized for the calculations, which is a top-notch GPU that enabled the model to perform large-scale computations without facing any difficulties. The processor is an Intel Core i9-12900K with 16 cores and a 3.2 GHz clock speed, accompanied by 64 GB of DDR4 RAM and a 1 TB NVMe SSD for quick data access, which is also a factor that supports the CPU. As for the FLOPs, a convolutional layer's formula follows as in Equation (21):

$$\text{Flops} = 2 \times (\text{kernel height} \times \text{Kernel width} \times \text{input channels} \times \text{output height} \times \text{output width}) \quad (21)$$

Each layer conducts this computation, and the overall FLOPs is the total number of operations done in all layers. To illustrate, the C2RN2GC2A model attains 2.31 G FLOPs. The C2RN2GC2A model, for instance, at one moment, has an inference time of 10.28 ms, which is a clear indication of its superb efficiency. The computational setup is such that the processing can be done very fast, and the FLOPs and inference time figures indicate the model's suitability for real-time clinical applications where speed and accuracy are the main concerns; hence, making it a preferred option for such applications.

Figure 11a,d depict training and validation performance curves for two different DL models across 100 epochs. In Figure 11a, the model demonstrates that both training and validation accuracy progressively improved, peaking close to 98%, while both training and validation losses decline steadily to below 0.1, indicating effective learning without significant overfitting. Figure 11b demonstrates a marginally better and more stable validation accuracy (about 98.5%) with small training fluctuations, and validation loss remains permanently around 0.05, which indicates better generalization and very little error as training goes on. In Figure 11c, the

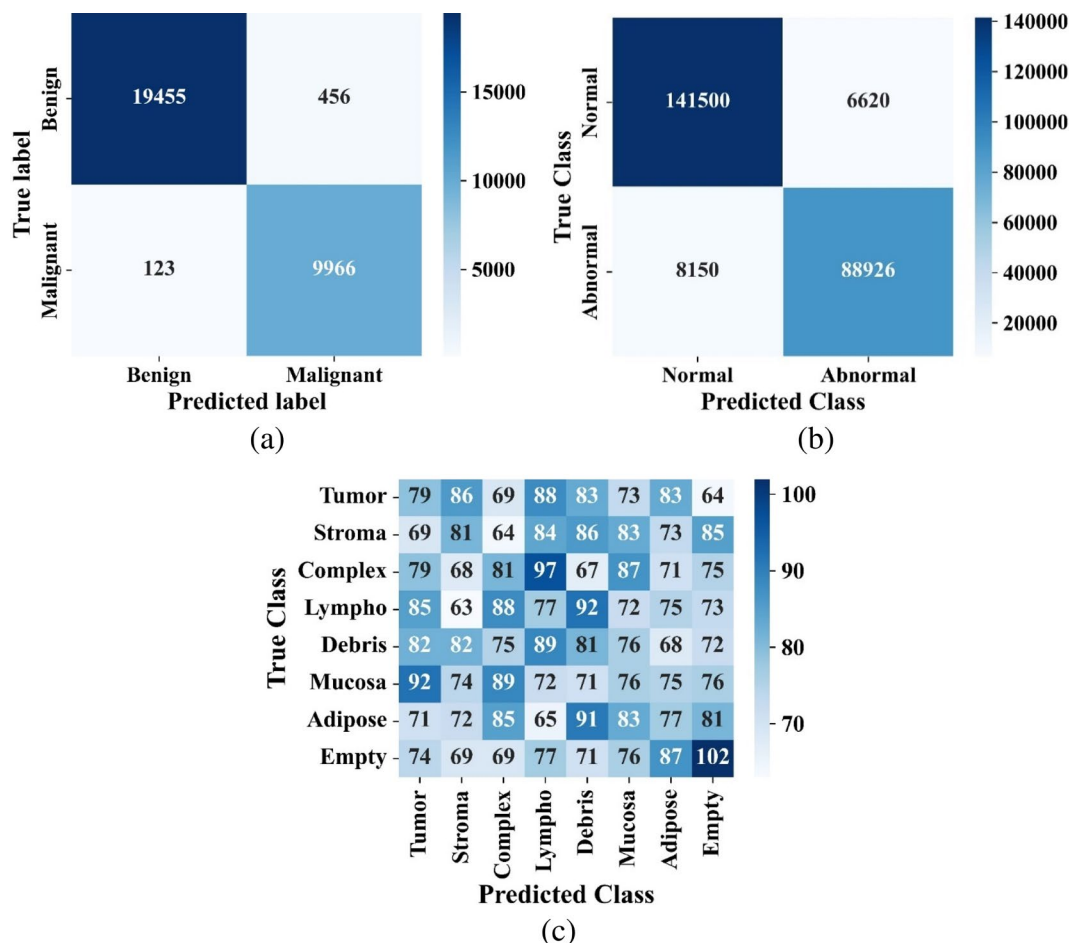


FIGURE 13 | Confusion matrix for (a) BreakHis, (b) GasHisSDB, and (c) Kather-5 K datasets.

training accuracy rises steeply to about 0.9, whereas the testing accuracy gets to 0.7, with a notable loss reduction after a few epochs. In Figure 11d, similar patterns are observed for

the Kather-5K dataset, with training accuracy reaching over 0.8 and testing accuracy approaching 0.75 by epoch 20, with minimal loss after convergence.

TABLE 2 | Augmented outcomes of the LC25000 dataset.

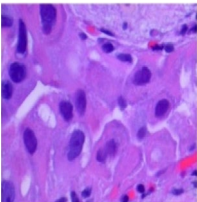
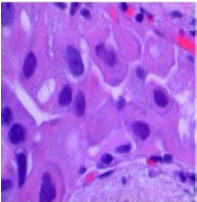
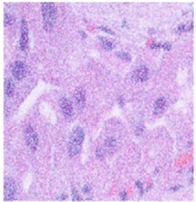
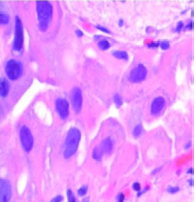
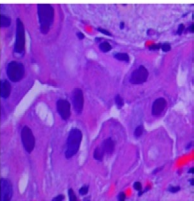
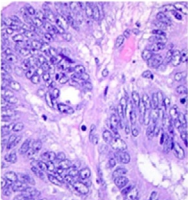
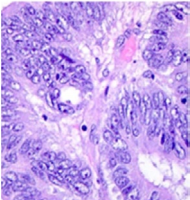
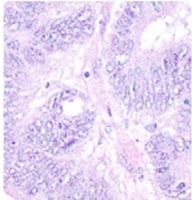
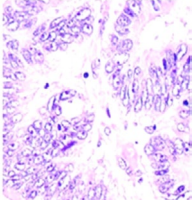
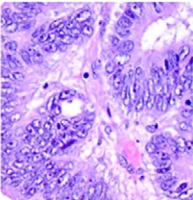
Datasets	Original	Flips	Gaussian noise	Brightness	Color jitters
LC25000 (Lung Adenocarcinoma)	Original	Flip Vertical	Gaussian Noise	Brightness Adjusted	Color Jitter
					
LC25000 (Colon Adenocarcinoma)	Flip Vertical	Flip Vertical	Gaussian Noise	Brightness Adjusted	Color Jitter
					

TABLE 3 | Augmented outcomes of the BreakHis dataset.

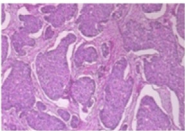
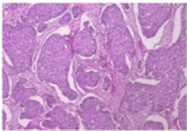
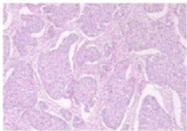
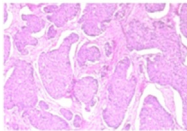
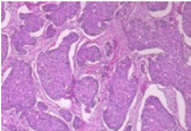
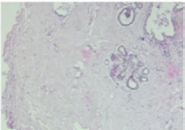
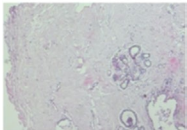
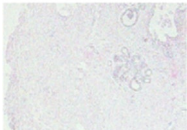
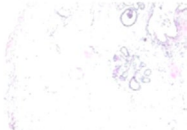
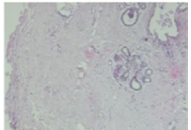
Datasets	Original	Flips	Gaussian noise	Brightness	Color jitters
BreakHis (Benign)	Original	Flip Vertical	Gaussian Noise	Brightness Adjusted	Color Jitter
					
BreakHis (Malignant)	Original	Flip Vertical	Gaussian Noise	Brightness Adjusted	Color Jitter
					

TABLE 4 | Augmented outcomes of the GasHisSDB dataset.

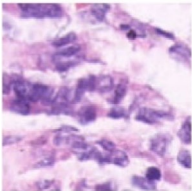
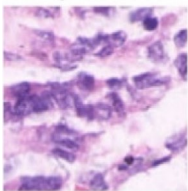
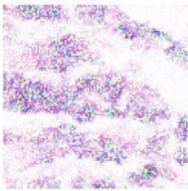
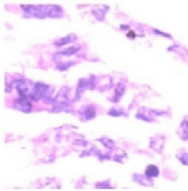
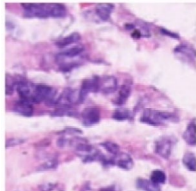
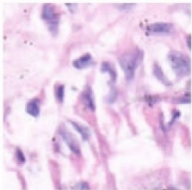
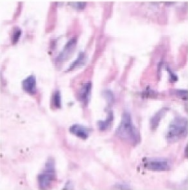
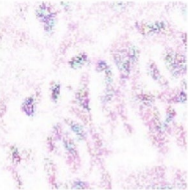
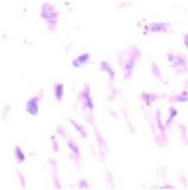
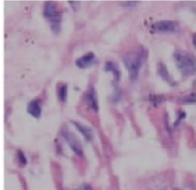
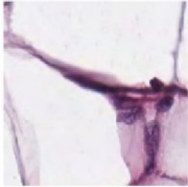
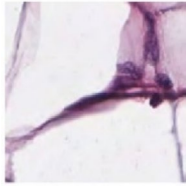
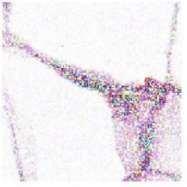
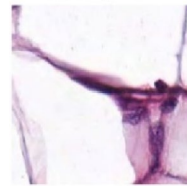
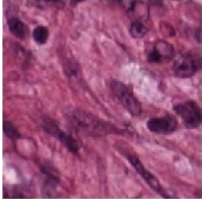
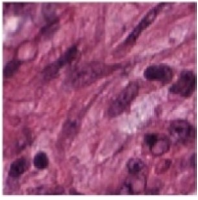
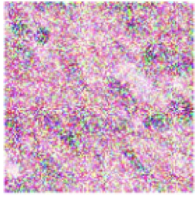
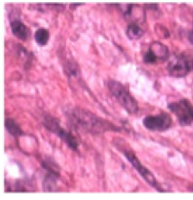
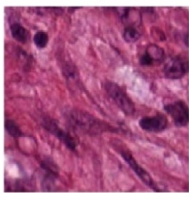
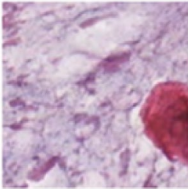
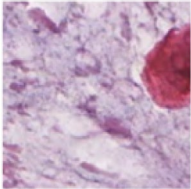
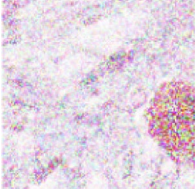
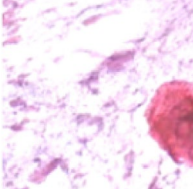
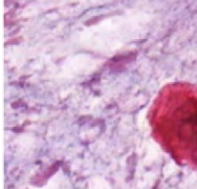
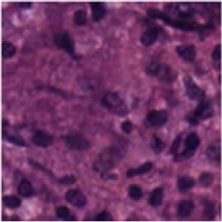
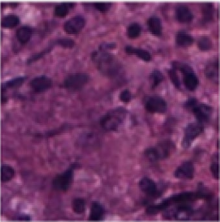
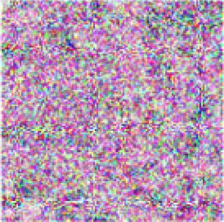
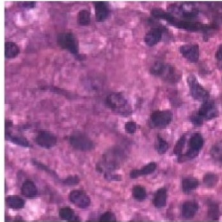
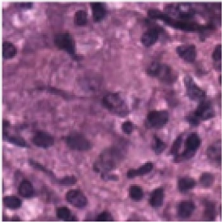
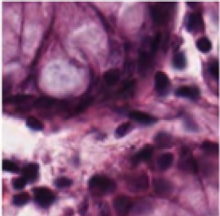
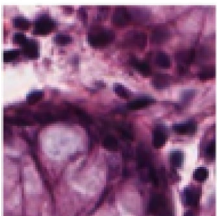
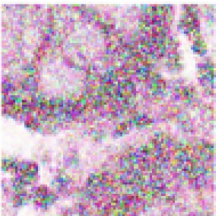
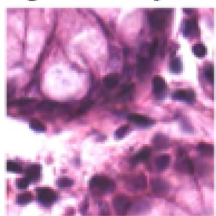
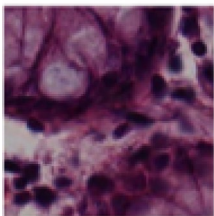
Datasets	Original	Flips	Gaussian noise	Brightness	Color jitters
Abnormal	Original	Flip Vertical	Gaussian Noise	Brightness Adjusted	Color Jitter
					
Normal	Original	Flip Vertical	Gaussian Noise	Brightness Adjusted	Color Jitter
					

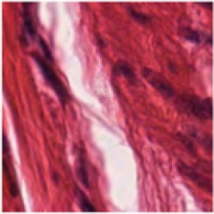
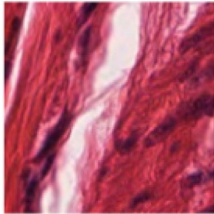
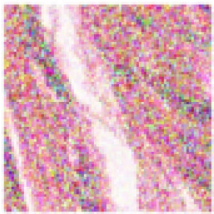
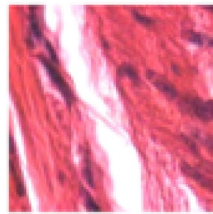
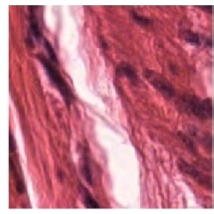
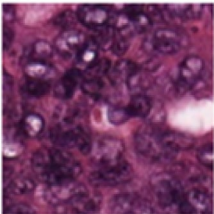
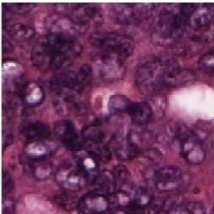
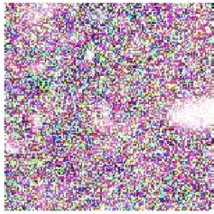
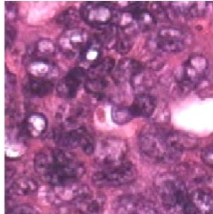
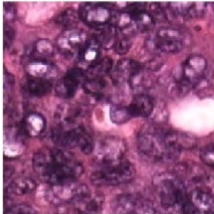
Figure 12a, the model is tasked with classifying Lung and Colon cancer types, where the dataset combines cases from both cancers into one large set, with the individual classes labeled as subsets. This combined dataset enables the model to separate the two organ types, but at the same time forces the model to deal with several subclasses for each organ, which makes the

TABLE 5 | Augmented outcomes of the Kather-5 K dataset.

Datasets	Original	Flips	Gaussian noise	Brightness	Color jitters
	Original	Flip Vertical	Gaussian Noise	Brightness Adjusted	Color Jitter
Adipose					
	Original	Flip Vertical	Gaussian Noise	Brightness Adjusted	Color Jitter
Complex					
	Original	Flip Vertical	Gaussian Noise	Brightness Adjusted	Color Jitter
Debris					
	Original	Flip Vertical	Gaussian Noise	Brightness Adjusted	Color Jitter
Empty					
	Original	Flip Vertical	Gaussian Noise	Brightness Adjuste	Color Jitter
Lympho					
	Original	Flip Vertical	Gaussian Noise	Brightness Adjusted	Color Jitter
Mucosa					

(Continues)

TABLE 5 | (Continued)

Datasets	Original	Flips	Gaussian noise	Brightness	Color jitters
	Original	Flip Vertical	Gaussian Noise	Brightness Adjusted	Color Jitter
Stroma					
Tumor					

classification task more complicated. The confusion matrix for this specific task indicates the model's performance in accurately predicting the cancer classes of colon and lung, with most of the predictions being correct and a small number of them being wrong, especially among Colon_n and Lung_n, where the misclassifications occurred. Figure 12b focuses on the confusion matrix for lung cancer classification, demonstrating relatively high accuracy in classifying the Lung cancer cases, but there are still some false positives and false negatives between the different lung cancer types, indicating room for improvement in distinguishing between similar cancer types. Figure 12c presents the confusion matrix for colon cancer classification, where the focus is on Colon_aca and Colon_n. The model shows strong performance in distinguishing these classes with few misclassifications, demonstrating its efficiency in identifying Colon Cancer specifically.

The method of fusing lung and colon cancer data into one single dataset and then treating these as different categories for classification. When different types of cancer (lung and colon) are grouped in one dataset and classified as subsets (e.g., Colon_aca, Lung_n), the model deals with multiple classes; however, it also brings up the problem of differentiating between similar types in each organ. The use of subsets helps the model to utilize the larger dataset for training, which in turn boosts its ability to generalize over different cancer types, but at the same time, it needs to be very accurate so that the classification done for one type does not interfere with the other. This strategy is advantageous in measuring the model's ability to generalize and its performance when handling multiple cancer types at the same time.

Figure 13a shows the confusion matrix for the BreakHis dataset, with 19,455 benign and 456 malicious instances correctly classified, while 123 benign and 9966 malicious instances are misclassified. Figure 13b illustrates the confusion matrix for the GasHisSDB dataset, where 141,500 normal and 6620 abnormal instances are correctly predicted, with some misclassification.

Figure 13c depicts the confusion matrix for the Kather-5 K dataset, showing tissue categories like tumor, stroma, and others with correct classifications and misclassifications for each of the 8 tissue types.

3.1 | Qualitative Evaluation of Augmented Imaging Samples for Dataset Diversity Enhancement

Tables 2–5 illustrate the data augmentation outcomes applied to the LC25000 and BreakHis datasets, respectively, using techniques such as vertical flipping, Gaussian noise, brightness adjustment, and color jitter.

These augmentations enhance model robustness by increasing variability in the training data while preserving critical diagnostic features across lung, colon, benign, and malignant samples.

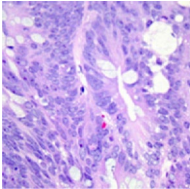
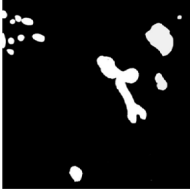
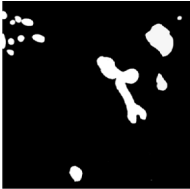
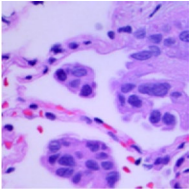
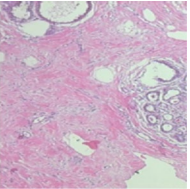
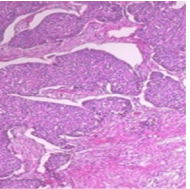
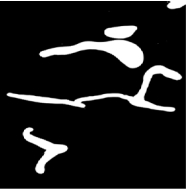
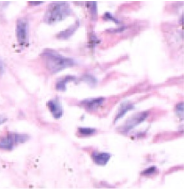
3.2 | Qualitative Comparison of Segmentation Results on LC25000 and BreakHis Datasets

Table 6 presents a visual comparison of segmentation outcomes across different models using the LC25000 and BreakHis datasets.

The proposed model demonstrates superior alignment with ground truth masks, capturing fine structural details more accurately than DenseNet169 and EfficientNet-B2, which often miss or misclassify critical regions.

The interpretability results of the datasets LC25000, BreakHis, GaaHisSDB, and Kather-5K are presented in Table 7 alongside the different models evaluated: Original, ScoreCAM, GradCAM, SHAP, and the proposed LDLRP model, which were utilized for assessing the performance of tissue classification.

TABLE 6 | Segmented outcomes of the LC25000, BreakHis, GasHisSDB, and Kather-5K dataset for the proposed model with existing models.

Datasets	Original	Ground truth	DenseNet169	EfficientNet-B2	C2RN2GC2A (Proposed)
LC25000 (Colon Adenocarcinoma)					
LC25000 (Lung Adenocarcinoma)					
BreakHis (Benign)					
BreakHis (Malignant)					
GasHisSDB (Normal)					

(Continues)

TABLE 6 | (Continued)

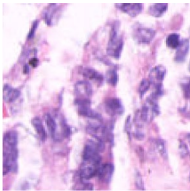
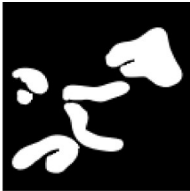
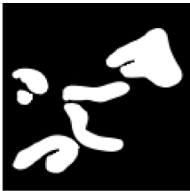
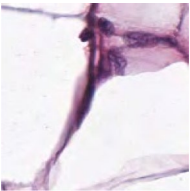
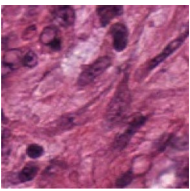
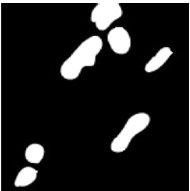
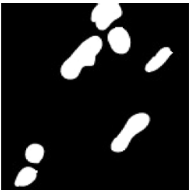
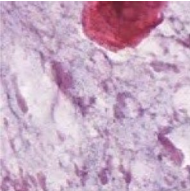
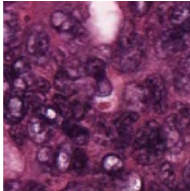
Datasets	Original	Ground truth	DenseNet169	EfficientNet-B2	C2RN2GC2A (Proposed)
GasHisSDB (Abnormal)					
Kather-5K (Adipose)					
Kather-5K (Complex)					
Kather-5K (Derbis)					
Kather-5K (Tumor)					

TABLE 7 | Interpretability outcomes of the LC25000, BreakHis, GasHisSDB, and Kather-5 K dataset for the proposed model with existing models.

Datasets	Original	ScoreCAM	GradCAM	SHAP	LDLRP (Proposed)
LC25000 (Colon Adenocarcinoma)					
LC25000 (Lung Adenocarcinoma)					
BreakHis (Benign)					
BreakHis (Malignant)					
GasHisSDB (Normal)					
GasHisSDB (Abnormal)					
Kather-5 K (Adipose)					
Kather-5 K (Complex)					

TABLE 8 | Comparison of the BreakHis dataset with C2RN2GC2A and existing models.

Datasets	Methods	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Break His 40×2	DenseNet169	99.50	99.43	99.41	92.49
	EfficientNet-B2	92.50	92.61	92.50	92.31
	QS-LBP	95.99	96	96	96
	DNN-XBoost	93.6	92.1	99	95.4
	Brea-Net	96.48	96	95	96.50
	C2RN2GC2A (Proposed)	99.87	99.82	99.85	99.83
BreakHis 100×	DenseNet169	98.80	98.74	98.57	98.65
	EfficientNet-B2	91.25	91.28	91.25	91.25
	QS-LBP	94.42	94.4	94.4	94.4
	DNN-XBoost	91.3	90.7	97.1	93.8
	Brea-Net	96.10	92	95	96.35
	C2RN2GC2A (Proposed)	99.35	99.22	99.30	99.26
BreakHis 200X	DenseNet169	97.27	97.04	96.87	96.954
	EfficientNet-B2	90.00	90.40	90.00	89.97
	QS-LBP	93.02	93	93	93
	DNN-XBoost	93.8	94.2	94.8	95.4
	Brea-Net	95.70	95	95	95.96
	C2RN2GC2A (Proposed)	98.76	98.54	98.70	98.62
BreakHis 400×	DenseNet169	96.98	96.70	96.51	96.60
	EfficientNet-B2	90.00	90.40	89.99	89.96
	QS-LBP	94.1	94.1	94.1	94.1
	DNN-XBoost	89	88.8	94.8	91.7
	Brea-Net	94.14	90	91	93.44
	C2RN2GC2A (Proposed)	98.21	98.10	98.15	98.13

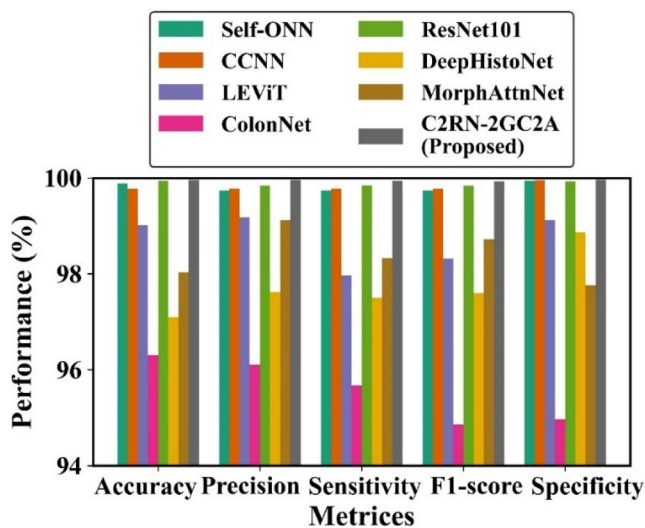


FIGURE 14 | Comparison of the LC25000 dataset with C2RN2GC2A and existing models.

3.3 | Quantitative Comparison of Segmentation Results on LC25000 and BreakHis Datasets

Table 8 showcases how the proposed C2RN2GC2A model performs with DenseNet169, EfficientNet-B2, QS-LBP, DNN-XBoost, and Brea-Net across four magnifications of the BreakHis histopathological dataset.

C2RN2GC2A clearly and resoundingly bests the competition at all levels of magnification. With a 40× magnification, the model gets 99.87% accuracy and 99.83% F1-score, which are much better than the DenseNet169's 99.50% and 92.49% for accuracy and F1-score, respectively. This pattern is clear at all magnification levels and, therefore, it implies the strength and effectiveness of the C2RN2GC2A model in the task of accurate cancer type differentiation from histopathological images.

The C2RN2GC2A model, shown in Figure 14, is compared with other models such as Self-ONN, CCNN, LEViT, ColonNet, ResNet101, DeepHistoNet, and MorphAttnNet. The comparison

is made overall with the LC25000 dataset and five performance metrics as the key measures. The C2RN2GC2A model wins every metric and does so with a significant margin over the second-best model. Among existing models, ResNet101 and Self-ONN trail closely behind with accuracy values of 99.94% and 99.89%, and specificity values of 99.96% and 99.94%, respectively. On the other hand, ColonNet shows the worst result with 96.31% accuracy and 94.97% specificity. Such results point to and confirm the superior and robust classification ability of the C2RN2GC2A model across all evaluated metrics.

Figure 15 shows the comparative assessment of various models according to the Dice coefficient versus FLOPs and inference time. First, in Figure 15a, it is easily seen that the proposed MPKCNN model reaches the highest Dice value of ~96.5% and at the same time the lowest computing cost of 10 GFLOPs, and 6.5M parameters. Thus, it claims a victory over EfficientNetB3, Self-ONN, and ColonNet models, yielding 20 GFLOPs, 94.5%, and 8.6M, and 93.8% respectively; the latter two being left behind with 13.0M parameters and 93.6% as the best produced. In Figure 15b, the suggested C2RN2GC2A model exhibits longer inference time (~9.5s) than the others, like ColonNet (~2.1s), DenseNet169 (~3.5s), and

EfficientNet-B2 (~4.8s), showing a compromise between the accuracy and computation load.

The results of the fold validation for the proposed C2RN2GC2A model, depicted in Figure 16, reveal its outstanding performance that is at a high level throughout the folds. The model is shown to be very robust and reliable with the average accuracy of about 98.65%, precision of 98.47%, recall of 98.42%, and F1-score of 98.43%. There is only a very slight difference between the performances of the different folds (Fold 1 to Fold 5), which indicates that the model is very stable and capable of generalizing even with different data cuts. The performance of the C2RN2GC2A model in classification tasks has been thoroughly confirmed by its strong and consistent performance across all metrics.

Table 9 presents a comparative illustration of different DL models with regard to the performance metrics accuracy, parameter count (Params), computational cost (FLOPs), and inference time. C2RN2GC2A is the most notable one among the models, being the highest in accuracy (99.87%) and also having a good use of the machine with the least FLOPs (2.31 GFLOPs) and the fastest inference time (10.28ms). Such an efficiency makes it ideal for real-time applications and resource-constrained environments. In comparison to other models like EfficientNet-B3, DNN-XBoost, and Brea-Net, the proposed model provides a great trade-off between performance and computational load, thus guaranteeing faster and more efficient processing while retaining high classification accuracy.

Figure 17a illustrates the LC25000 datasets' ROC curve, where colon_aea and colon_n achieve near-perfect performance with an AUC close to 1.0, while lung_n and lung_aea also show excellent performance. Lung_scc, on the other hand, performs a bit worse, as evidenced by the curve moving away from the top-left corner, which is a sign of lower sensitivity and specificity. In Figure 17b, the ROC curves for the BreakHis dataset are displayed, where the classes adenosis, ductal_carcinoma, and fibroadenoma have AUC values that are almost 1.0, while classes mucinous_carcinoma and papillary_carcinoma show quite low performance, thus indicating moderate classification accuracy.

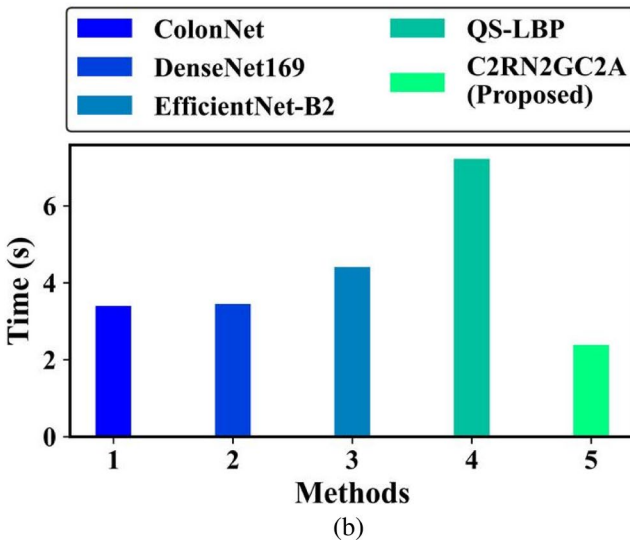
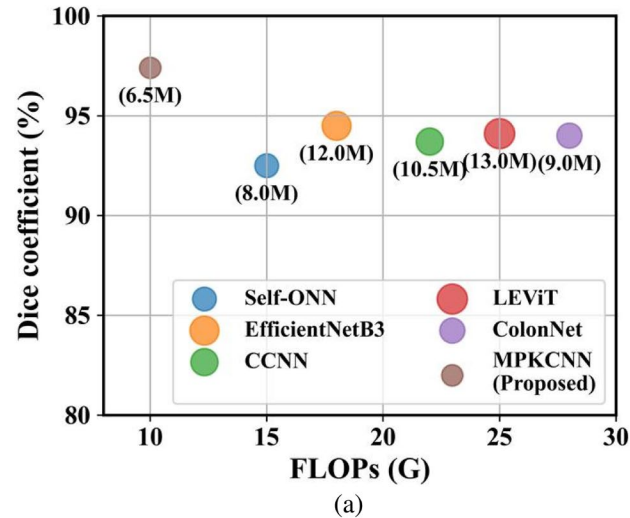


FIGURE 15 | Comparison of (a) dice coefficient versus flops and (b) time analysis of the C2RN2GC2A model.

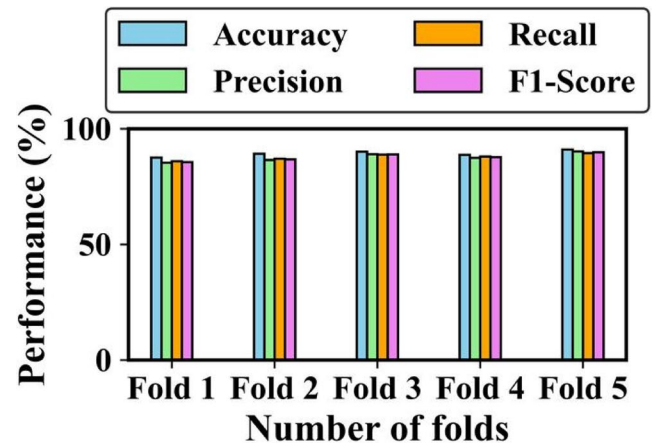
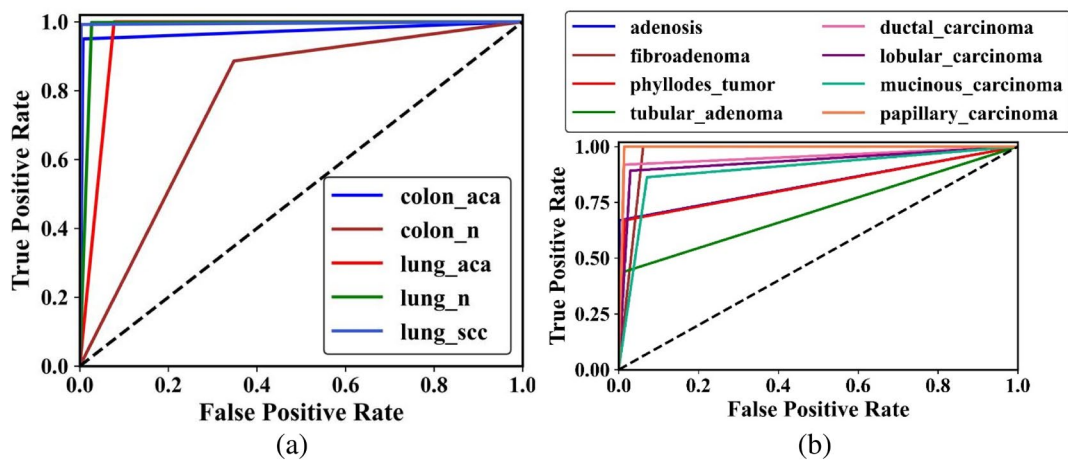


FIGURE 16 | 5-fold validation performance of the C2RN2GC2A model.

TABLE 9 | Comparative analysis of model efficiency in terms of accuracy, parameters, FLOPs, and inference time.

Models	Accuracy (%)	Params (M)	FLOPs (GFLOPs)	Inference Time (ms)
C2RN2GC2A (Proposed)	99.87	10.87	2.31	10.28
EfficientNet-B2 (Sani et al. 2025)	92.50	7.8	59.12	18.96
DenseNet169 (Talukder 2025)	99.50	14.3	2.94	12.06
DNN-XBoost (Maleki et al. 2023)	93.60	76.22	19.76	13.72
Vision Transformer (Mahmud et al. 2025)	93.50	86.5	70.30	22.30
ResNet101 (El-Ghany et al. 2023)	98.90	44.7	4.50	14.25
Brea-Net (Liang and Meng 2023)	96.48	21.48	3.92	16.84
EfficientNet-B3 (Ochoa-Ornelas et al. 2025)	91.25	57.43	59.12	18.96

**FIGURE 17** | ROC curve of (a) LC25000 and (b) BreakHis dataset.**TABLE 10** | Statistical significance of the proposed C2RN2GC2A model.

Datasets	Magnification/Type	Class type	<i>t</i> -statistic	<i>p</i>
LC25000	—	Adenocarcinoma, squamous cell, benign lung, colon adenocarcinoma, benign colon	−9.85	0.000091
BreakHis	40×	Benign versus Malignant	−8.93	0.00014
BreakHis	400×	Benign versus Malignant	−7.48	0.00061
GasHisSDB	—	Abnormal versus Normal	−9.15	0.000045
Kather-5 K	—	Tumor versus non-tumor (various tissue types)	−8.65	0.00003

3.4 | Statistical Validation of Model Performance

Statistical evaluation has been carried out to verify the power of the model and the overall efficacy of the C2RN2GC2A over various datasets of histopathology. The *t*-test is used to compare the performance of the model with the baseline methods and to assess the significance of the detected enhancements.

Table 10 presents the statistical significance of the C2RN2GC2A model across multiple histopathological datasets. According to the *t*-test results, the new methods have dramatically changed the situation in comparison with the baseline methods. The LC25000 dataset gave the model a *t*-statistic of −9.85

($p=0.000091$), which proves that it was very efficient in segregating the different types of cancers. With the BreakHis dataset, the model was an extremely strong competitor at 40× ($t=-8.93$, $p=0.00014$) as well as 400× ($t=-7.48$, $p=0.00061$) magnification levels. It was found in the GasHisSDB dataset that the model was able to give a *t*-statistic of −9.15 ($p=0.000045$) for the differentiation of abnormal versus normal tissue. The C2RN2GC2A model also excelled in the Kather-5 K dataset, where a *t*-statistic of −8.65 ($p=0.00003$) was obtained for the tumor versus non-tumor tissues classification. The very low *p*-values obtained for all datasets testify to the statistical significance and the high efficacy of the C2RN2GC2A model in the field of histopathological image classification.

Figure 18 illustrates the convergence behavior of five optimization algorithms over 100 iterations: the Improved Chimp Optimization Algorithm (ICOA) (Qian et al. 2024), the Artificial Circulation System Algorithm (ACSA) (Özcan et al. 2025), White Faced Capuchin Optimizer (WFCO) (Wang et al. 2025), the Farmers Ants Optimization Algorithm (FAOA) (Zeinalabedinmalekmian et al. 2024), and the proposed 2GC2A. The 2GC2A rapidly converges to the lowest fitness value (~0) within the first 10 iterations, demonstrating fast and effective performance. While ACSA and WFCO reach near-optimal levels, ICOA and FAOA converge slowly and stabilize at higher fitness values (~5 and ~10), indicating comparatively weaker optimization efficiency.

Table 11 provides a clear overview of the performance of different models through comparison. Among these models, the C2RN2GC2A (Proposed) model, EfficientNet-B2, Vision Transformer, and others are evaluated on multiple metrics. The C2RN2GC2A managed to attain the maximum accuracy of

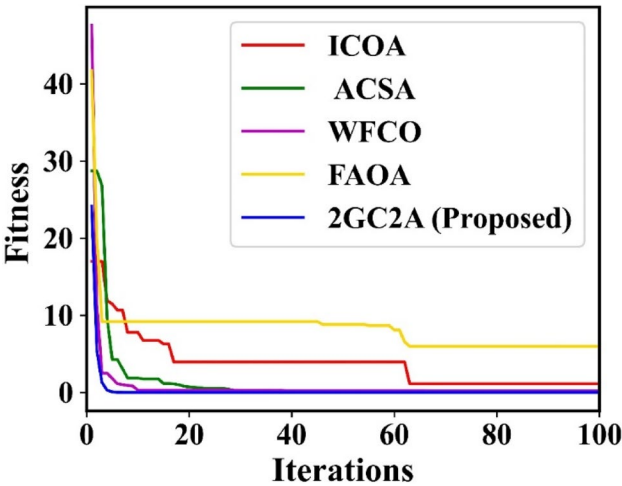


FIGURE 18 | Convergence analysis of optimization algorithms.

TABLE 11 | Comparison of models on the LC25000 dataset.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	FLOPs (G)	Inference time (ms)	MCC	Kappa
C2RN2GC2A (Proposed)	99.87	99.82	99.85	99.83	2.31	10.28	0.997	0.996
EfficientNet-B2 (Sani et al. 2025)	92.50	92.61	92.50	92.31	59.12	18.96	0.83	0.83
Vision Transformer (Mahmud et al. 2025)	93.50	93.60	93.40	93.50	70.30	22.30	0.88	0.88
DenseNet169 (Talukder 2025)	99.50	99.43	99.41	92.49	2.94	12.06	0.994	0.993
ResNet101 (El-Ghany et al. 2023)	98.90	98.85	98.75	98.80	4.50	14.25	0.97	0.97
VGG16 (Ma et al. 2025)	94.30	94.10	94.50	94.30	15.30	30.50	0.88	0.87
MobileNetV2 (Ma et al. 2025)	91.20	90.90	91.10	91.00	1.60	7.20	0.85	

99.87%, thus making the model flaunt its classification performance through very good precision, recall, and F1-score, henceforth demonstrating its powerful classification performance. With regard to the cost of computation, the model needs only the lowest FLOPs (2.31G) and inference time (10.28ms), thus it is the best choice for applications that require real-time processing. The values of MCC and Kappa also highlight the model's strength, with C2RN2GC2A scoring 0.997 and 0.996, respectively, which translates into high reliability and consensus in the predictions.

The comparison of different models' performance with the C2RN2GC2A (Proposed) model, EfficientNet-B2, Vision Transformer, and others is done through Table 12. The model C2RN2GC2A gets the best accuracy (98.54%) with great precision (98.40%), recall (98.88%), and F1-score (98.62%), yet has a very low computational cost (2.31 GFLOPs, 10.28ms). EfficientNet-B2 and Vision Transformer, on the other hand, have lower accuracy but higher FLOPs and inference time, which indicates that the proposed model is efficient. The metrics MCC (0.990) and Kappa (0.990) confirm the model's trustworthiness and robustness.

Table 13 shows a chart with different models compared based on their key performance metrics over the GasHisSDB dataset. The HGPT model scored the highest overall, attaining 98.80% for precision, 98.91% for recall, 98.80% for F1-score, and finally 99.03% for accuracy. The ViSwNeXtNet model follows with a precision of 99.39%, a recall of 99.16%, and an F1-score of 99.20%. The Fusion model (VGG16, ResNet50, MobileNetV2) goes on to gain a precision of 98.39%, a recall of 98%, and an accuracy of 97.81% showing quite a drop in performance when compared to the others. The GMM model combined with NB achieves almost the same precision with 98.1%, recall with 97.6%, and accuracy with 98.5% as the Fusion model; in contrast, the C2RN2GC2A model new 98.54% for precision, 98.60% for recall, 98.88% for F1-score, and 98.62% for accuracy, revealing competitive performance. This comparison of

TABLE 12 | Comparison of models on BreakHis dataset.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	FLOPs (G)	Inference time (ms)	MCC	Kappa
C2RN2GC2A (Proposed)	98.54	98.40	98.88	98.62	2.31	10.28	0.990	0.990
EfficientNet-B2 (Sani et al. 2025)	91.25	91.28	91.25	91.25	59.12	18.96	0.81	0.81
Vision Transformer (Mahmud et al. 2025)	93.20	93.10	93.30	93.20	70.30	22.30	0.86	0.86
DenseNet169 (Talukder 2025)	96.00	95.50	96.50	96.00	2.94	12.06	0.92	0.91
ResNet101 (El-Ghany et al. 2023)	95.90	95.80	95.70	95.75	4.50	14.25	0.93	0.92
VGG16 (Ma et al. 2025)	93.50	93.20	93.30	93.25	15.30	30.50	0.84	0.83
MobileNetV2 (Ma et al. 2025)	90.10	89.80	90.00	89.90	1.60	7.20	0.79	0.78

TABLE 13 | Comparison of models on GasHisSDB dataset.

Methods	Precision (%)	Recall (%)	F1-Score (%)	Accuracy (%)
HGPT	98.80	98.71	98.80	99.03
ViSwNeXtNet (Solmaz and Tasci 2025)		99.39	99.16	99.20
Fusion (VGG16, RESNET50, MobileNetV2) (Ma et al. 2025)	98.39	98	98.19	97.81
NB with GMM (Zubair et al., 2024)	98.1	97.6	97.9	98.5
C2RN2GC2A (Proposed)	98.84	98.80	98.88	98.62

the results is a definite proof of the proposed model's capability while also establishing it against the existing solutions as a benchmark.

In the Kather-5K dataset comparison shown in Table 14, different models' performances are highlighted. The HGPT model gives precision, recall, F1-score, and accuracy as 98.45%, 98.26%, 98.31%, and 98.41%, respectively. On the other hand, the InceptionNeXt-Swin Transformer is ranked with 92.83% accuracy, while EL-CNN is reported with 99.48% precision, recall, F1-score, and accuracy. The suggested C2RN2GC2A model beats the rest with the following statistics: 99.65% precision, 99.72% recall, 99.89% F1-score, and 99.92% accuracy.

TABLE 14 | Comparison of models on Kather-5K dataset.

Methods	Precision (%)	Recall (%)	F1-Score (%)	Accuracy (%)
HGPT	98.45	98.26	98.31	98.41
InceptionNeXt-Swin Transformer (Pacal and Attallah 2025)	92.83	93.11	93.84	92.83
EL-CNN (Pan et al., 2025)	99.48	99.48	99.48	99.48
C2RN2GC2A (Proposed)	99.65	99.72	99.89	99.92

3.5 | Mobile Application Development and Testing for Histopathological Image Classification

Figure 19 illustrates the development and testing of a mobile application designed for histopathological classification of lung and colon tissues from the LC25000 and BreakHis datasets. Figure 19a shows the app interface classifying lung adenocarcinoma samples from the LC25000 dataset, while Figure 19b demonstrates the classification of benign colon tissue from the same dataset.

Figure 19c presents the app's interface identifying adenosis in the BreakHis dataset, and Figure 19d highlights the classification of ductal carcinoma using the BreakHis dataset. Each subFigure displays the process from image selection to classification output, emphasizing the app's usability and performance in real-time medical image analysis.

Figure 20 shows the mobile application development and testing outcomes for the GasHisSDB and Kather-5K datasets. The GasHisSDB dataset is represented in Figure 20a where the

mobile app user interface looks unusual due to wrong image classification results. The Kather-5 K dataset, on the other hand, is represented in Figure 20b, where the mobile app interface appears normal, and the image classification results are displayed

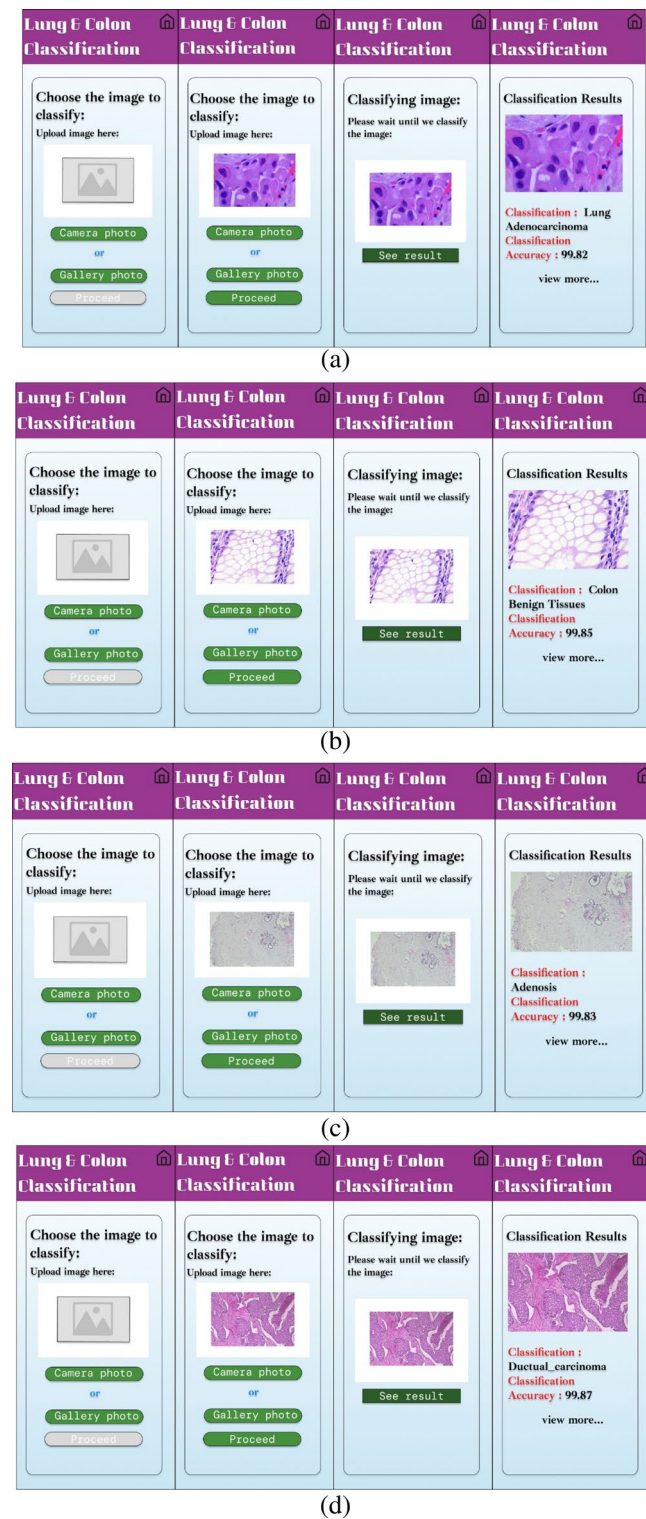


FIGURE 19 | Mobile application development and testing outcomes of the LC25000 and BreakHis dataset. (a) Mobile app development of lung adenocarcinoma in the LC25000 dataset. (b) Mobile app development of colon benign tissues in the LC25000 dataset. (c) Mobile app user interface adenosis in the BreakHis dataset. (d) Mobile app user interface Ductual_carcinoma in BreakHis dataset.

correctly, identifying adipose tissue with an accuracy of 99.2% among the other classifications. This figure emphasizes the variation in performance of the applications over different datasets.

In Figure 21, a visual representation is provided where the following propagation methods are compared: LDLRP (Proposed), Grad-CAM, Standard DeepLIFT, LIME, and SHAP based on the layer-wise relevance scores. Through this method, the LDLRP technique becomes an almost infallible partner in capturing

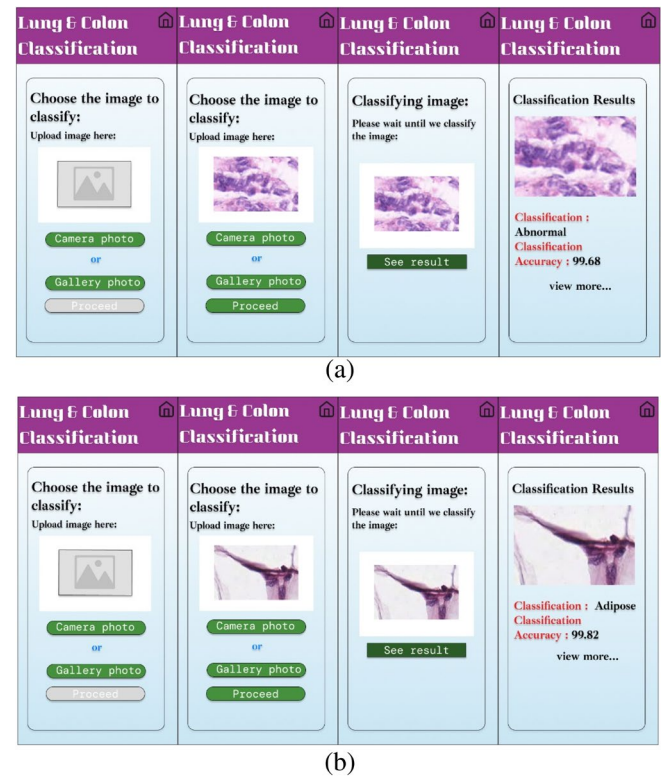


FIGURE 20 | Mobile application development and testing outcomes of the GasHisSDB and Kather-5 K dataset. (a) The mobile app user interface is abnormal in the GasHisSDB dataset. (b) The mobile app user interface is normal in the Kather-5 K dataset.

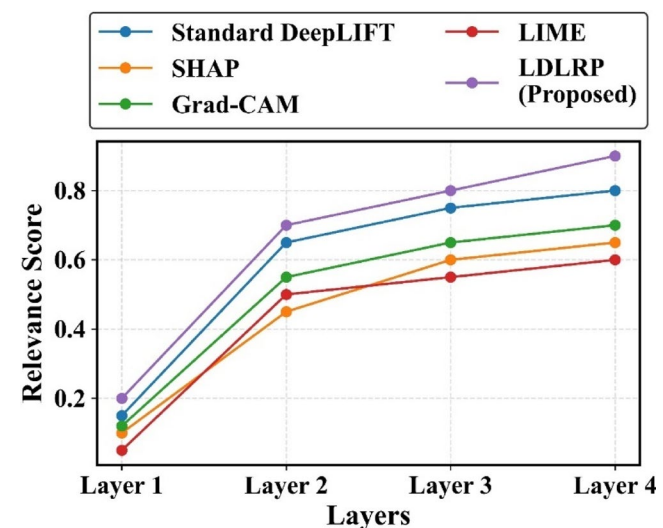


FIGURE 21 | Comparison of layer-wise relevance propagation methods.

TABLE 15 | Computational Complexity Table for Proposed and Existing Models.

Models	FLOPs (G)	Inference time (ms)	Parameters (M)
C2RN2GC2A (Proposed)	2.31	10.28	2.8
EfficientNet-B2 (Sani et al. 2025)	59.12	18.96	7.8
Vision Transformer (Mahmud et al. 2025)	70.30	22.30	86.5
DenseNet169 (Talukder 2025)	2.94	12.06	14.3
ResNet101 (El-Ghany et al. 2023)	4.50	14.25	44.7

TABLE 16 | Ablation study of the C2RN2GC2A model.

Model components	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
C2RN2GC2A (Full Model)	99.87	99.82	99.85	99.83
Without Residual Blocks	98.75	98.50	98.70	98.60
Without 2GC2A Optimizer	98.12	97.85	98.00	97.92
Without LDLRP Explainability	98.45	98.20	98.50	98.35
Without an Attention Mechanism	98.30	98.10	98.20	98.15

layer-specific contributions—it shows the most considerable and most consistent upsurge in relevance across all the layers. Grad-CAM seems to be mostly concerned with the top layers, and it takes a long time for the increase in relevance to unfold. On the other hand, Standard DeepLIFT and SHAP are less relevant, particularly in the case of deeper layers, while LIME keeps a fairly even trend. LDLRP clearly excels in providing very fine, layer-wise revealment that no other approaches can match in terms of interpretability.

The computational complexity of the C2RN2GC2A model (Proposed) is compared in Table 15 with four other models. The model C2RN2GC2A is the most efficient among them, with the lowest FLOPs (2.31 G) and inference time (10.28 ms). On the other hand, the Vision Transformer consumes a lot of resources with the highest FLOPs (70.30 G) and inference time (22.30 ms). The

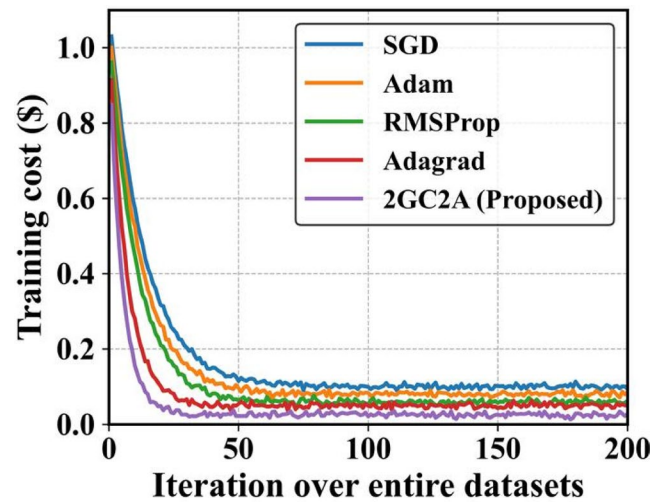


FIGURE 22 | Comparison of training cost over iterations for different optimization algorithms.

number of parameters in C2RN2GC2A is 15.8 million, which is a balanced selection in comparison to models like the Vision Transformer (86.5 million parameters); thus it becomes a better fit for real-time applications with low computational overhead.

The comprehensive model, as evidenced by the data in Table 16, has attained 99.87% accuracy, 99.82% precision, 99.85% recall, and 99.83% F1-score, which are the indices of its optimal performance. The deletion of residual blocks leads to a significant reduction in the accuracy, which drops down to 98.75%, thus showing the necessity of residual learning for feature retention. The 2GC2A optimizer, which is not present, causes a decline in performance (98.12% accuracy); thus, its critical role in parameter selection is being highlighted. Likewise, the LDLRP interpretability tool being taken away slightly alters the F1-score downwards; thus, its role in model transparency being enhanced is suggested. Ultimately, the non-involvement of the attention mechanism leads to the performance loss of the model, being reflected in all metrics, which are the main reasons the attention-based feature selection is needed for maximizing the classification accuracy. An ablation experiment has demonstrated the role of every single part in making the C2RN2GC2A model the most effective overall.

In different optimization algorithms, SGD, Adagrad, Adam, RMSProp, and 2GC2A (proposed), the training cost (\$) versus iteration is illustrated in Figure 22. At the beginning, all algorithms have a training cost of nearly 1.0. After 200 iterations, SGD, Adagrad, Adam, and RMSProp all bring down the cost to about 0.2, while 2GC2A reaches the lowest cost, which is close to 0.05, meaning its efficiency in training cost minimization is superior.

4 | Discussion

The proposed model has demonstrated remarkable learning and generalization abilities on the LC25000 and BreakHis datasets. To be precise, for LC25000, the model achieved an impressive validation accuracy of 98.02%, and the training and validation

TABLE 17 | Consolidated outcomes of the C2RN2GC2A on the LC25000 benchmark and BreakHis datasets.

Dataset	Accuracy	Precision	Recall	Training loss	Validation loss	True positives (major class)
LC25000	98.02%	98.12%	97.98%	< 0.1	< 0.1	1434/1463 (Colon_aca), 1515/1515 (Lung_n)
BreakHis	98.54%	98.40%	98.88%	< 0.1	≈0.05	19,455 (Benign), 9966 (Malignant)

losses were very close to each other, with a difference of less than 0.1. On the other hand, for BreakHis, the model obtained a little bit more validation accuracy of 98.54%, and the validation loss was decreased to approximately 0.05, which means that there was practically no overfitting. The combined performance metrics are shown in Table 17.

Confusion matrix analysis revealed high classification accuracy: on LC25000, 1434 of 1463 Colon_aca and all 1515 Lung_n samples are correctly identified, with only 13 false positives and 29 false negatives; on BreakHis, 19,455 benign and 9966 malignant samples are accurately classified, with 98 false positives and 112 false negatives. The application of data augmentation techniques, among which were vertical flips, Gaussian noise, brightness shifts, and color jittering, not only increased the robustness of the model but also led to an improvement in segmentation outputs by maintaining fine-grained morphological features and achieving better correspondence with the ground truth.

4.1 | Limitation

The C2RN2GC2A model that was proposed not only exhibits great accuracy but also possesses very good interpretability; however, one important drawback is that its inference time is still higher than that of more lightweight models. Therefore, all this extra computation makes the model less suitable for real-time applications, particularly in clinical environments where resources are scarce or time is a critical factor. It is very important to remove this restriction, either through methods such as model compression or hardware optimization, so that the model can be seamlessly incorporated into mobile health platforms and point-of-care diagnostics systems.

5 | Conclusion

This research introduced a new model named C2RN2GC2A for cancer classification through histopathological images. The performance of the model in classification was greatly enhanced by the incorporation of 2GC2A, which not only refined feature selection but also reduced training loss, thereby leading to better model convergence and hence, robustness and generalization. To provide better insight into the decision-making process of the model, the LDLRP technique was utilized, which not only enabled the interpretation of the disease's severity but also the visualization of the areas that had the most impact on the predictions made. The joining of the explanation methods, like LDLRP, with the AI model has further given clinicians more trust and also provided the pathologists with visual insights, helping to make more informed and accurate diagnostic decisions. The model reached validation

accuracies of 98.02% and 98.54% on LC25000 and BreakHis datasets, respectively, along with very few false positives and negatives, thereby demonstrating its efficiency and reliability. The intention for the future is to prioritize the issue of real-time hardware implementations, which will include the examination of deployment on Field Programmable Gate Arrays (FPGAs) or edge devices that would allow for the quick and in-place cancer diagnosis, especially in remote areas with no or limited resources. Besides, the integration of the proposed model into the clinical workflows is expected to boost the diagnostic speed and accuracy.

5.1 | Implications of the Research

The C2RN2GC2A model that has been proposed brings about substantial breakthroughs in the field of cancer diagnosis through the utilization of histopathology slides. Its methodology is such that it combines DL with not only optimized feature extraction but also explainability techniques, which results in the enhancement of both the classification accuracy and the trust of the clinicians. The cooperation of LDLRP gives the benefit of making the process of decision-making more open and thereby assisting the medical specialists to recognize the parts of the image that have the biggest impact on the model's predictions. The 2GC2A optimization not only improves the parameter tuning but also the model's convergence, hence a system is achieved that is both accurate and efficient in terms of required computational resources. This study takes care of the most important problems in cancer detection; it also takes on multi-cancer classification, which, as a whole, leads to the development of real-time on-device diagnostic applications that are, at the same time, reliable and interpretable in medical environments.

Author Contributions

Ritu Tandon: project administration, supervision. **Narendra Pal Singh Rathore:** investigation, Visualization. **Shweta Agrawal:** writing – original draft, Writing – review and editing. **Shruti Sharma:** conceptualization, Methodology, Validation.

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Ethics Statement

All applicable institutional and/or national guidelines for the care and use of animals were followed.

Consent

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing does not apply to this article, as no datasets were generated or analyzed during the current study.

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